

Research paper

Numerical investigation of the impact of hydraulic behavior of various constrictions on a Venturi channel

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ARTICLE INFO

Keywords:

Venturi channel
Constriction geometry
Turbulent kinetic energy
Energy dissipation
Computational fluid dynamics

ABSTRACT

This study comprehensively investigates the hydraulic behavior of Venturi channels, which are engineered with a deliberate reduction in cross-sectional area to accelerate fluid velocity, under varying constriction geometries. The research aims to understand their impact on flow rate, turbulence, and energy dissipation. Employing both experimental data and numerical simulations via computational fluid dynamics (CFD), three constriction ratios $b/B = 0.35, 0.50$, and 0.65 (where b is constriction width and B is channel width) were analyzed. The analysis specifically evaluated flow patterns, velocity profiles, turbulent kinetic energy (TKE), and energy dissipation. Results reveal that narrower constrictions (e.g., $b/B = 0.35$) significantly increase turbulence and create steeper velocity gradients, leading to prolonged recovery lengths and elevated energy dissipation, particularly at lower flow rates. Conversely, wider constrictions (e.g., $b/B = 0.65$) facilitate smoother flow transitions and consistently demonstrate lower energy dissipation across all flow conditions. The energy dissipation analysis highlights a critical trade-off between turbulence intensity and flow stability, where narrower configurations improve energy efficiency at higher flow rates but incur greater energy losses at lower flow rates. These findings underscore the substantial impact of constriction geometry on optimizing Venturi channels for diverse engineering applications, including fluid measurement, wastewater treatment, and hydropower systems. The study's outcomes provide a robust foundation for enhancing the efficiency and design of Venturi channels under various operational conditions.

1. Introduction

Venturi channels, with their distinctive converging-diverging geometry, play a vital role in various applications requiring efficient control of fluid flow. The narrowing, or constriction, within these channels accelerates the fluid and introduces unique flow characteristics, making them invaluable in industrial processes and environmental systems. This narrowing creates a critical flow regime that can influence phenomena such as bubble formation, flow measurement accuracy, and overall hydraulic performance. Several studies have focused on evaluating the hydraulic efficiency and flow rate estimation in Venturi channels. For instance, Zuikov and Bakunyaeva [1] proposed a method for calculating flume flow rate with high accuracy, benchmarking it against the Russian Federation State Standard. Chhantyal et al. [2] developed mechanistic models for mass flow estimation in open Venturi channels, though these models often require fluid rheology-dependent tuning correction factors, limiting their applicability without real-time rheological measurements.

Beyond flow rate estimation, the complex flow dynamics within Venturi channels have been extensively explored, often leveraging computational fluid dynamics (CFD). Recent CFD investigations have provided significant insights, particularly concerning two-phase flow phenomena and geometric influences. Zhao et al. [3] utilized high-speed imaging to analyze bubble movement and breakup in Venturi channels with varying divergent angles, emphasizing the impact of geometry. Similarly, Liu et al. [4] employed experimental analysis to examine cavity behavior and wall static pressure under different divergence angles. Sun et al. [5] designed three curved flumes based on critical-flow theory and validated them through experiments and simulations (max error 4.7%). Results showed reduced head loss (up to 38.4% at $CR=0.4$) and improved discharge accuracy, making the design suitable for flat and mild-slope channels. Huang et al. [6] further explored Venturi channels as efficient bubble generators, highlighting how their geometry creates high turbulence and shear flow beneficial for bubble formation, with performance largely dependent on geometric parameters. More specifically, CFD has been instrumental in detailing bubble dynamics and cavitation. Ding et al. [7,8] employed the Volume of Fluid

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<https://doi.org/10.1016/j.rineng.2025.106916>

Received 19 May 2025; Received in revised form 30 July 2025; Accepted 22 August 2025

Available online 25 August 2025

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Abbreviations and Numenclature

b	Constriction width of the Venturi channel (m)
B	Main channel width (m)
b/B	Constriction ratio (-)
L	Length of the Venturi channel (m)
Q	Flow discharge or flow rate (m ³ /s)
h ₁	Upstream flow depth (m)
E ₁	Upstream water head (m)
E ₂	Downstream water head (m)
V ₁	Upstream flow velocity (m/s)
m	Discharge coefficient (-)
$\Delta E/E_1$	Energy dissipation ratio
E_2/E_1	Residual energy ratios
Re	Reynolds number (-)
Fr ₁	Upstream Froude number (-)
ρ	Fluid density (kg/m ³)

P	Pressure (Pa)
u, v, w	Velocity components in x, y, z directions (m/s)
A _x , A _y , A _z	Fractional areas open to flow in x, y, z directions
(G _x , G _y , G _z)	Mass accelerations in x, y, z directions
(f _x , f _y , f _z)	Viscous accelerations in x, y, z directions
r	Refinement ratio (ratio of coarse to fine cell size)
Cell size/B	Dimensionless cell size relative to channel width
TKE	Turbulent kinetic energy (J/kg)
TI	Turbulence intensity (%)
TD	turbulence dissipation (J/kg/s)
CFD	Computational Fluid Dynamics
VOF	Volume of Fluid method
RNG k- ϵ model	Renormalization Group k-epsilon turbulence model
LES	Large Eddy Simulation
SST k- ω model	Shear Stress Transport k-omega turbulence model
APE	Absolute Percentage Error (%)

(VOF) model coupled with the level-set method to simulate virtual bubble motion and deformation in steady Venturi flows, providing detailed insights into breakup characteristics, trajectory evolution, and the positive correlation between bubble breakup and turbulence within the diverging section. Their findings revealed that increasing the diverging angle (β) reduces the average bubble diameter while enhancing the pressure drop, whereas increasing the throat diameter (D) produces the opposite effect. Similarly, Wang et al. [9] utilized the VOF model to investigate velocity fields and flow distribution in an innovative porous Venturi configuration, aiming to optimize bubble generation and mass transfer efficiency. Gou et al. [10] advanced this line of research by exploring cavitation intensity in Venturi geometries using Large Eddy Simulation (LES) in conjunction with the Zwart cavitation model, elucidating flow dynamics under expanding channel conditions. In a complementary study, Lecardonnell et al. [11] underscored the complexities of accurately modeling liquid–vapor phase distributions, emphasizing the continued necessity of experimental validation due to inherent modeling challenges. Süme [12] conducted both numerical and experimental investigations on the impact of abrupt, non-continuous contractions on supercritical flow energy dissipation, employing CFD with RNG turbulence modeling. Experiments using trapezoidal and semicircular contractions (10–15 cm) across Froude numbers ranging from 2.5 to 7 demonstrated that energy loss increased with rising Froude numbers, reaching up to 69.5 % greater than that observed in a free hydraulic jump for a 15 cm contraction. Patricio et al. [13] demonstrated the Venturi effect on photoinjected hole recombination, identifying a non-uniform diffusion velocity profile characteristic of viscous hydrodynamic behavior.

Despite numerous studies on Venturi channels, a significant research gap persists regarding the detailed hydraulic behavior of single-phase flow under varying constriction geometries, particularly in terms of turbulent kinetic energy (TKE) distribution and energy dissipation. While many CFD-based investigations have focused on multiphase flow phenomena such as cavitation and bubble formation, the systematic analysis of how different constriction ratios influence key flow parameters like velocity profiles, turbulence characteristics, and energy dissipation remains underexplored. This gap limits the ability to optimize Venturi channel designs for applications that demand precise control over flow dynamics and energy efficiency, especially beyond traditional two-phase systems. Given the crucial role of constricted regions in shaping flow behavior through acceleration, pressure variation, and turbulence generation, a deeper understanding of these effects in relation to constriction ratio is essential. Accordingly, the present study aims to address this knowledge gap by conducting a comprehensive investigation of the hydraulic behavior in open Venturi channels with varying

constriction ratios ($b/B = 0.35, 0.50$, and 0.65). Using a combination of experimental measurements and computational fluid dynamics (CFD) simulations, this research focuses on: (i) analyzing velocity profiles and flow patterns within the constricted sections, (ii) examining the spatial distribution of turbulence intensity and TKE, (iii) evaluating energy dissipation across different geometries, and (iv) providing insights for optimizing constriction designs to improve efficiency and performance in diverse engineering and environmental applications. The findings are expected to bridge the existing gap in understanding the link between constriction geometry and energy-related flow behavior, thereby contributing valuable guidance for advanced Venturi channel design.

2. Materials and methods

2.1. Experimental setup

This study adapts the configuration of a Venturi channel to serve as a comparative framework for investigating flow behavior and constriction effects. The setup is based on prior laboratory experiments described by Zuikov and Bakunyaeva [1] in Fig. 1. The Venturi channel employed in this research is 15.5 m long, with a throat width of 0.153 m, a height of 0.45 m, and a main channel width of 0.31 m Table 1. The channel walls are made of tempered glass, and the end plates are constructed from stainless steel, with a roughness coefficient (Manning's n) of 0.009. The gradient settings of the channel are adjustable using a GSZ-100 jack system, allowing slope variations of ± 0.0175 . Water flow measurement is facilitated by an electromagnetic Promag 10D flowmeter with a 0.3 % accuracy rating. Additional digital sensors measure water depth with a 0–455 mm range and a precision of 0.01 mm. The hydraulic behavior is analyzed using the Bernoulli equation, which considers energy conservation between sections. In this configuration, the flow transitions through critical depth within the throat, with minimal hydraulic losses assumed. Moreover, the hydraulic characteristics of the flow are shown in Table 1.

2.2. Numerical models

Numerical simulations were performed using Computational Fluid Dynamics (CFD), based on the experimental studies of Venturi channel flow behavior conducted by Zuikov and Bakunyaeva [1]. The setup utilized precise dimensions and configurations, including a throat width of 0.153 m, a main channel width of 0.31 m, and a height of 0.45 m. The simulations aimed to validate the experimental data and assess flow characteristics, including the transitions through critical depth, the effects of constriction, and the discharge efficiency under varying flow

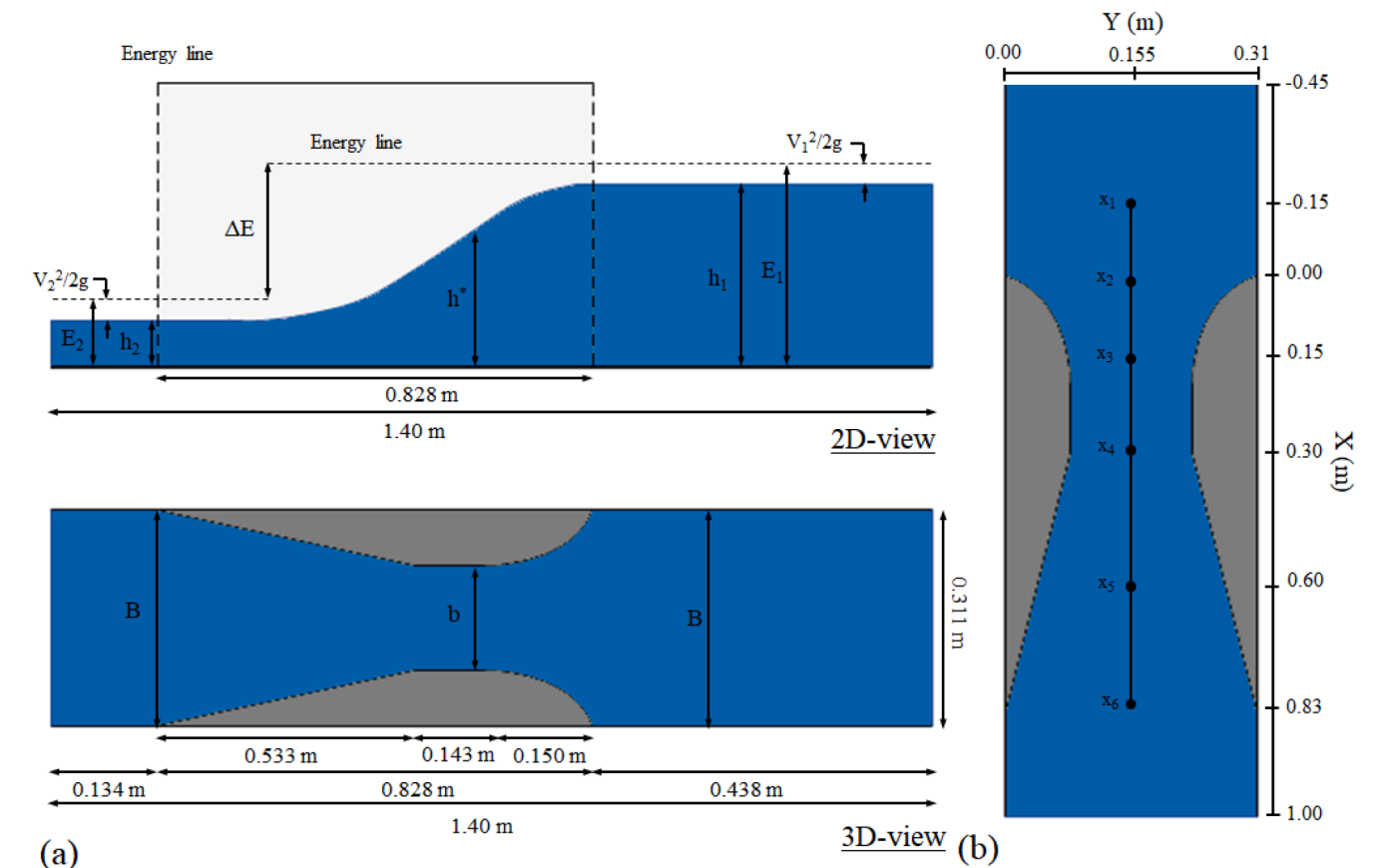


Fig. 1. a) Schematic diagram of the flow-measuring Venturi channel; b) Selected point from channel.

Table 1
Studies of the hydraulic characteristics of the flow-measuring Venturi channel.

Model	Q (m ³ /s)	h ₁ (m)	E ₁ (m)	V ₁ (m/s)	m	Re	Fr ₁	B (m)	b (m)	b/B (m)	L (m)
VC 0.50	0.031	0.228	0.238	0.4315	0.204	78.928	0.289	0.31	0.15	0.5	0.83
	0.027	0.207	0.216	0.4106	0.203	69.896	0.287				
	0.023	0.185	0.194	0.3945	0.202	60.223	0.285				
	0.018	0.162	0.169	0.3583	0.201	50.025	0.284				
	0.014	0.135	0.141	0.3301	0.201	39.12	0.285				
	0.01	0.107	0.112	0.2921	0.2	28.298	0.283				
	0.006	0.073	0.077	0.2423	0.201	16.976	0.285				
	0.001	0.029	0.031	0.161	0.206	4741	0.291				

* V₁ is upstream velocity, h₁ is upstream flow depth, m is the discharge coefficient, Fr₁ and Re are upstream Froude number and Reynolds number and E₁ is upstream water head.

Table 2
Hydraulic characteristics of the flow-measuring Venturi Channel.

Model	Q (m ³ /s)	h ₁ (m)	E ₁ (m)	V ₁ (m/s)	m	Re	Fr ₁	B (m)	b (m)	b/B (m)	L (m)
VC 0.35	0.031	0.184	0.199	0.42	0.142	77.298	0.313	0.31	0.1	0.35	0.83
	0.026	0.173	0.186	0.399	0.14	68.941	0.306				
	0.023	0.153	0.165	0.375	0.14	57.436	0.306				
	0.018	0.132	0.142	0.351	0.135	46.332	0.308				
	0.014	0.11	0.119	0.324	0.135	35.684	0.312				
	0.01	0.088	0.094	0.295	0.134	25.96	0.318				
	0.006	0.06	0.065	0.254	0.133	15.264	0.332				
	0.002	0.025	0.027	0.182	0.127	4552	0.367				
VC 0.65	0.031	0.283	0.29	0.382	0.249	108.219	0.230	0.31	0.2	0.65	0.83
	0.026	0.26	0.266	0.343	0.243	89.258	0.215				
	0.023	0.235	0.24	0.331	0.245	77.691	0.218				
	0.018	0.206	0.21	0.312	0.245	64.272	0.219				
	0.014	0.174	0.177	0.275	0.244	47.867	0.211				
	0.01	0.137	0.14	0.242	0.245	33.168	0.209				
	0.006	0.095	0.097	0.203	0.244	19.295	0.210				
	0.002	0.04	0.041	0.131	0.227	5236	0.209				

conditions. Moreover, we investigate Venturi channel models with different constriction ratios, but identical overall dimensions, which are shown in Table 2 and Fig. 2. The constriction ratios are defined as $b/B = 0.35$, $b/B = 0.50$, and $b/B = 0.65$, where B is the channel width, b is the constriction width, and L is the Venturi length. The specifications of the models and the dimensions of the constrictions are provided in Table 3. Additionally, a sample 2D and 3D representation of the channel is depicted in Fig. 1. Following the validation of the model, the effect of eight different upstream flow rates on the flow behavior within the constriction regions will be analyzed and the hydraulic characteristics of the flow of these two constrictions are brought in Table 2.

2.3. Governing equations

The CFD software solves the Navier-Stokes–Reynolds equations for three-dimensional incompressible flow analysis using the fluid-volume method on a gridded domain. It incorporates Volume of Fluid the True VOF algorithm [14] to track free surface flow, defining fluid configurations through the VOF function $F(x, y, z, t)$. Here, F represents the fluid fraction, where $F = 1$ in fluid-filled cells and $F = 0$ in empty cells [15].

$$\frac{\partial(uA_x)}{\partial x} + \frac{\partial(vA_y)}{\partial y} + \frac{\partial(wA_z)}{\partial z} = 0 \quad (1)$$

where A_x , A_y , and A_z are fractional areas open to flow in the directions (x, y, z) , u , v , and w is velocity components in the $-x$, $-y$, and $-z$ directions. The Navier-Stokes equations with velocity components as the momentum equations define three-dimensional fluid mobility for Cartesian coordinates [16,17]:

$$\frac{\partial u}{\partial t} + \frac{1}{V_F} \left(uA_x \frac{\partial u}{\partial x} + vA_y \frac{\partial u}{\partial y} + wA_z \frac{\partial u}{\partial z} \right) = -\frac{1}{\rho} \frac{\partial P}{\partial x} + G_x + f_x \quad (2)$$

$$\frac{\partial v}{\partial t} + \frac{1}{V_F} \left(uA_x \frac{\partial v}{\partial x} + vA_y \frac{\partial v}{\partial y} + wA_z \frac{\partial v}{\partial z} \right) = -\frac{1}{\rho} \frac{\partial P}{\partial y} + G_y + f_y \quad (3)$$

Table 3

Characteristics of the meshes tested in the convergence analysis.

Case	Small size cell (m)	Large size cell (m)	r	Cells count	Mesh Type	Cell size/B
1	0.008	0.01	1.25	601,602	Coarse	0.026
2	0.007	0.009	1.28	957,056	Medium	0.023
3	0.006	0.008	1.33	1276,392	Fine	0.019
4	0.0055	0.007	1.27	1973,145	Very Fine	0.018

$$\frac{\partial w}{\partial t} + \frac{1}{V_F} \left(uA_x \frac{\partial w}{\partial x} + vA_y \frac{\partial w}{\partial y} + wA_z \frac{\partial w}{\partial z} \right) = -\frac{1}{\rho} \frac{\partial P}{\partial w} + G_z + f_z \quad (4)$$

Here, (G_x, G_y, G_z) represent mass accelerations, while (f_x, f_y, f_z) denote viscous accelerations in the x , y , and z directions, respectively. ρ is the fluid density, VOF is the Volume of Fluid of the flow, and P is the pressure. The RNG k - ϵ turbulence model was used to simulate flow turbulence. This model includes an additional term that enhances accuracy in predicting rapidly strained and swirling flows [15].

$$\frac{\partial(\rho k)}{\partial t} + \frac{\partial(\rho k u_i)}{\partial x_i} = \frac{\partial}{\partial x_j} \left[\alpha_k \mu_{eff} \frac{\partial k}{\partial x_j} \right] + G_k - \rho \epsilon \quad (5)$$

$$\frac{\partial(\rho \epsilon)}{\partial t} + \frac{\partial(\rho \epsilon u_i)}{\partial x_i} = \frac{\partial}{\partial x_j} \left[\alpha_\epsilon \mu_{eff} \frac{\partial \epsilon}{\partial x_j} \right] + C_{1\epsilon} \frac{\epsilon}{k} G_k - C_{2\epsilon} \rho \frac{\epsilon^2}{k} \quad (6)$$

The RNG k - ϵ turbulence model was employed in this study to simulate flow behavior within the Venturi channel. This model is particularly well-suited for flows involving rapid strain and swirling, both of which are characteristic of the converging-diverging geometry of Venturi systems. Compared to the standard k - ϵ model, the RNG variant incorporates enhanced transport equations that improve its ability to predict turbulence generation and dissipation in regions of high curvature and varying strain rates features that are central to the constriction zones examined in this study. Although alternative models such as the SST k - ω offer improved predictions in scenarios involving strong adverse pressure gradients and flow separation, and LES (Large Eddy Simulation) provides high-fidelity resolution of turbulent structures,

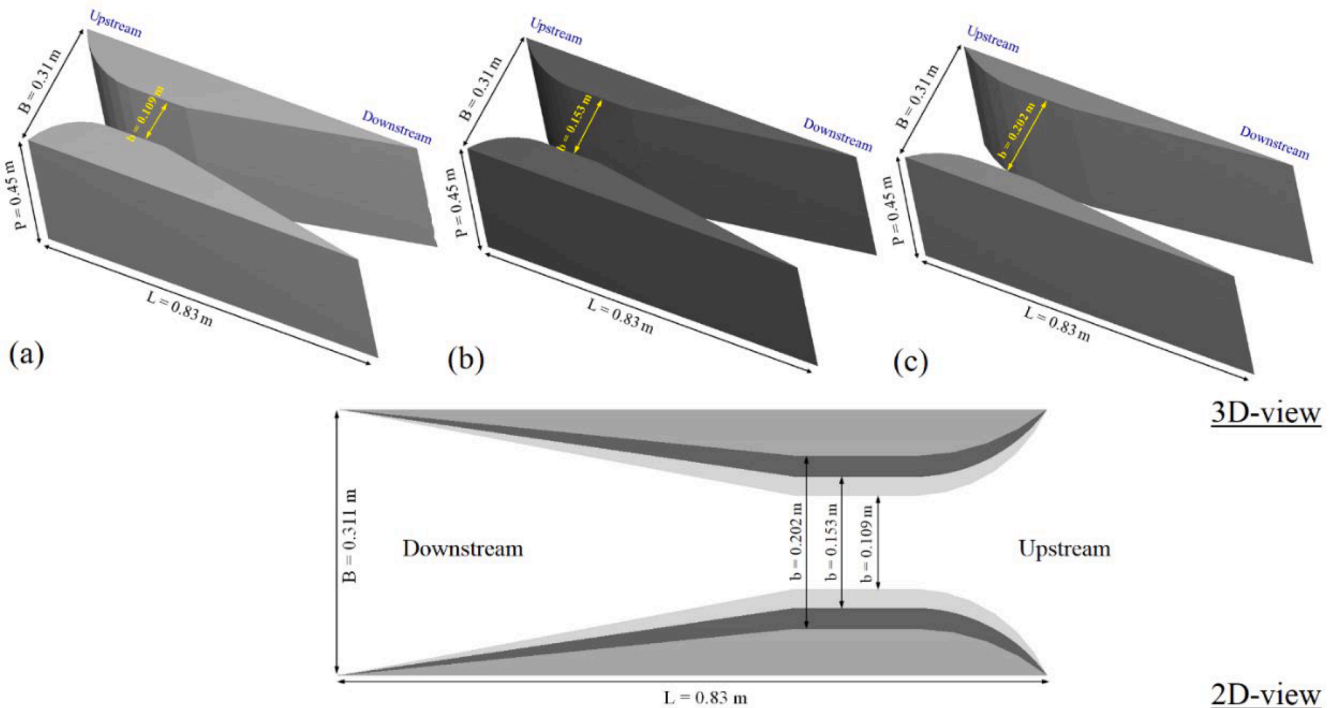


Fig. 2. Fig. 2. 2D and 3D Views of the Venturi with constriction widths $b = 0.109$ m, 0.153 m, and 0.202 m.

these approaches are either computationally intensive or more appropriate for unsteady flow regimes. Given the predominantly steady-state nature of the flow conditions investigated here and the focus on turbulent kinetic energy (TKE) and energy dissipation, the RNG $k-\epsilon$ model provided a robust balance between accuracy and computational efficiency. Previous studies including those by Sun et al. [5], Herrera-Granados [18], and Süme [12] have validated the effectiveness of the RNG $k-\epsilon$ model in simulating turbulence in channels with significant horizontal or vertical contractions. These findings support its selection for analyzing turbulence characteristics in geometries similar to the Venturi channel used here. Moreover, since large-scale flow separation was not a dominant feature in the scenarios considered, the RNG $k-\epsilon$ model was deemed sufficiently accurate for capturing the key flow dynamics and turbulence-related energy transformations essential to this study's objectives.

2.4. Computational domain, grid layout and the boundary conditions

Evaluating mesh sensitivity within the computational domain is critical to ensuring the accuracy and reliability of CFD simulations. An

optimal mesh is achieved when further refinement produces negligible changes in the solution, indicating convergence and numerical stability. In this study, the geometric domain of the Venturi channels was discretized using structured, rectangular grids. To assess mesh sensitivity, two mesh blocks with different resolutions were applied across the channel, with finer grids embedded within coarser ones in the constricted regions, forming a nested mesh configuration. The refinement ratio (r), defined as the ratio of coarse to fine cell sizes, ranged from 1.25 to 1.33 falling within the recommended range of 1.3 to 1.5 for reliable results, as suggested by Celik et al. [19] and summarized in Table 5. Four mesh configurations were tested for the teardrop-shaped Venturi channel to identify the most suitable grid arrangement. Numerical results, including upstream flow depth (h_1), velocity (V_1), and peak of the turbulent kinetic energy (TKE), were calculated and compared against laboratory data reported by Zuikov and Bakunyaeva [1] for a Venturi channel with $h_1/L = 0.22$ and flow rate $Q = 0.031 \text{ m}^3/\text{s}$. This validation ensured that the selected mesh configuration accurately captured key flow features while maintaining computational efficiency.

To calculate the flow velocity at a constant width of the Venturi channel, the following equation was used:

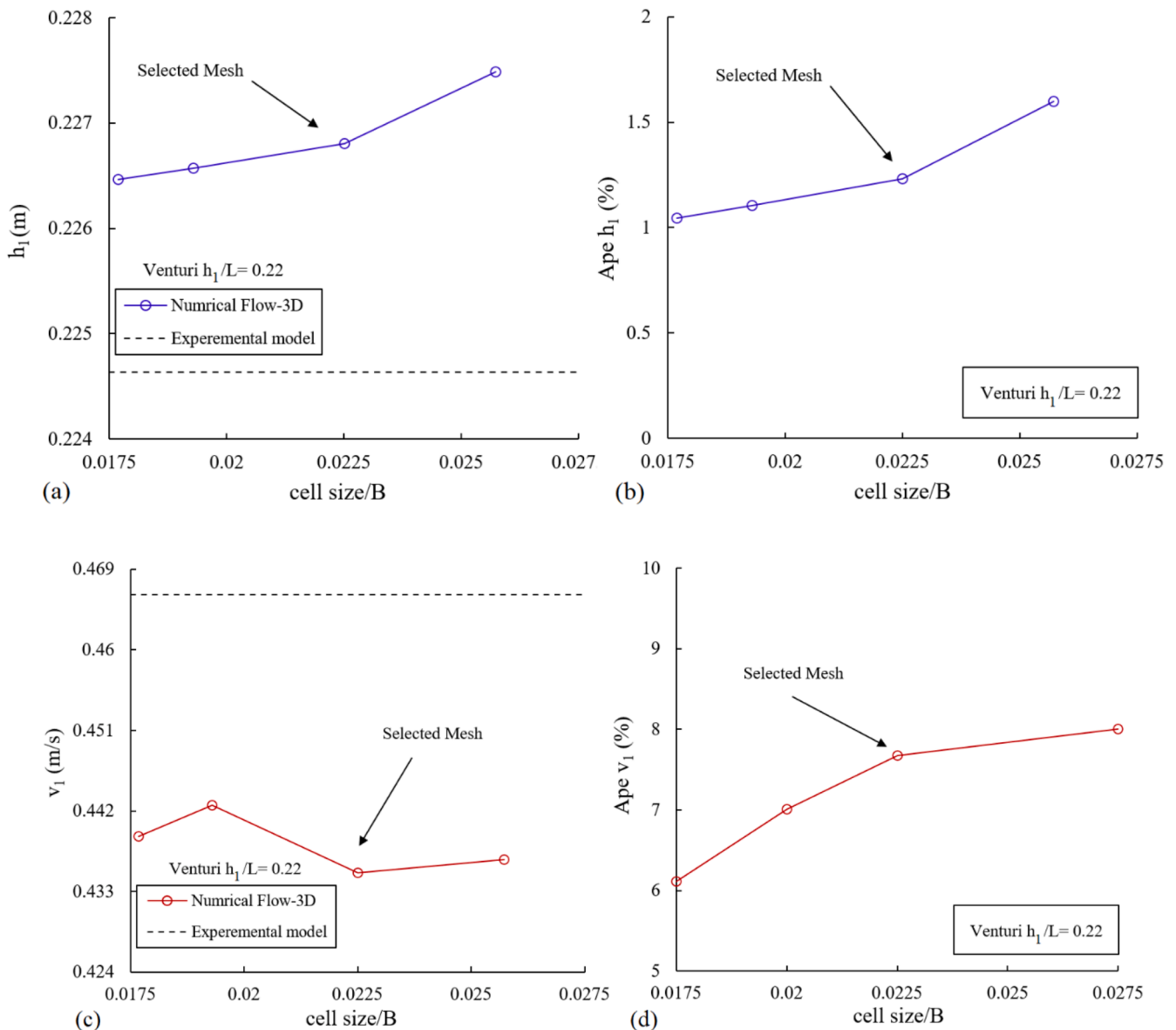


Fig. 3. Results of mesh sensitivity analysis for flow depth and flow velocity ($h_1/L = 0.22$).

$$V_1 = \frac{Q}{B * h_1} \quad (7)$$

where Q is the discharge through the channel, B is the channel width, and h_1 is the upstream flow depth. To evaluate the accuracy of the CFD model against experimental data, the absolute percentage error (APE) was calculated using Eq. (8):

$$APE = \frac{1}{N} \sum_{i=1}^N \left| \frac{M_{(EXP)} - M_{(NUM)}}{M_{(EXP)}} \times 100 \right| \quad (8)$$

Here, M_{NUM} and M_{EXP} represent the values of a given hydraulic parameter from numerical simulations and experimental measurements, respectively. Fig. 3 presents the comparison of upstream flow depth and velocity obtained from CFD simulations with experimental data, plotted against the dimensionless cell size (Cell size/ B), where $B = 0.311$ m. This comparison illustrates that the accuracy of the numerical model is strongly dependent on mesh resolution. As cell size decreases, model accuracy improves; however, below a certain threshold, further refinement yields negligible gains. According to the mesh sensitivity analysis, reducing the cell size below Cell size/ $B = 0.0225$ did not significantly enhance simulation accuracy. Therefore, this value was selected as the optimal mesh resolution. For the optimized mesh comprising 957,056 cells the absolute percentage errors for upstream depth and velocity were 6.67 % and 0.91 %, respectively. Table 6 summarizes APE values for V_1 and h_1 , along with the peak TKE observed at the channel throat ($x = 0$ m) across four mesh densities. Notably, the variation in peak TKE remained below 2 %, indicating convergence. This confirms that the selected mesh (Cell size/ $B = 0.0225$) provides a reliable balance between accuracy and computational efficiency, capturing both hydraulic behavior and turbulence characteristics effectively. Prior to addressing the main research objectives, a comprehensive mesh sensitivity analysis was performed. The final mesh configuration featured a primary cell size of 0.009 m, with a finer nested mesh of 0.008 m embedded in critical regions, particularly around the constriction. Further mesh refinement beyond this configuration did not result in measurable improvements in model accuracy, thereby validating the selected mesh as optimal for subsequent simulations.

Table 4

To ensure uniform flow conditions at the inlet boundary, the upstream distance was set to approximately twice the length of the constriction ($2L$, where L represents the length of the constricted region). The discharge rate (Q) and upstream flow depth (h_1) were applied as the inlet boundary conditions. At the downstream boundary, an outflow condition was imposed, ensuring no influence on the flow solution within the Venturi channel domain. The top boundary was set with an atmospheric pressure condition to model the free-surface flow, with a specified pressure of zero. The side walls and channel floor were modeled as no-slip boundaries, indicating that the velocity normal to these surfaces was zero. To address the interface between the coarse and fine mesh blocks within the constriction region, a symmetric boundary condition with a zero-gradient constraint was applied, ensuring smooth transitions and accurate representation of the flow dynamics [20]. These boundary conditions were selected based on the geometric and hydraulic characteristics of the Venturi channel models, as shown in Figs. 4

Table 4
Error values obtained from computational domain mesh sensitivity analysis.

Mesh Type	Cell size / B	Cells count	APE- V_1 (%)	APE- h_1 (%)	TKE (J/kg)	APE-TKE (%)
Coarse	0.0258	601,602	8.146	1.323	0.142	3.43
Medium	0.0225	957,056	7.678	1.02	0.146	2.82
Fine	0.0193	1276,392	7.006	0.911	0.148	1.37
Very Fine	0.0177	1973,145	6.11	0.866	0.149	0.68

and 5, to effectively simulate the impact of varying constriction ratios and flow rates on the channel's hydraulic performance.

Fig. 6 illustrates the convergence behavior of discharge values at three critical locations within the Venturi channel: upstream, downstream, and inside the constricted section, for a flow discharge of $Q = 0.031$ m³/s. The time-dependent flow (TDF) results demonstrate that the system requires approximately 10 s to reach a steady discharge state, after which fluctuations become negligible, indicating numerical stability and convergence of the solution. In fact, the flow discharge in all three sections stabilizes roughly within these 10 s. To ensure greater confidence in the results, a simulation duration of 20 s was employed for each run, allowing sufficient time for the flow to fully stabilize. To ensure the robustness of the simulation, the convergence criteria were carefully monitored through the temporal evolution of flow discharge at these three points, rather than relying solely on residual reduction metrics. This approach provides a direct physical measure of stability in the hydraulic parameters of interest. The steady-state solution is considered achieved when discharge variations fall below a predefined tolerance, confirming that the flow field has stabilized. In addition to discharge monitoring, the residuals for momentum, continuity, turbulent kinetic energy, and dissipation rate equations were tracked during the simulations, with convergence thresholds set to 10^{-6} , ensuring numerical precision and consistency. The time step was selected based on a Courant-Friedrichs-Lewy (CFL) number less than unity, balancing computational efficiency with solution accuracy. The nested mesh refinement strategy combined with the selected turbulence model (RNG $k-\epsilon$) further contributes to numerical stability by accurately resolving velocity gradients and turbulence characteristics within the constriction. The negligible differences in discharge values between upstream, downstream, and inside the constriction after the transient phase confirm that the numerical solution is both physically and mathematically converged. Overall, the convergence behavior demonstrated in Fig. 5, supported by stringent residual criteria and appropriate time-step control, validates the reliability of the numerical simulations in capturing the steady hydraulic performance of the Venturi channel.

2.5. Scale effects

Scale effects influence model tests and their upscaling to prototype scale. Highly turbulent air-water flows cannot be accurately modeled using Froude similitude alone due to the significant influence of viscosity and surface tension [21]. For a true similarity in the aeration process between the model and prototype, it is essential to simultaneously satisfy the Froude criterion. The dimensionless numbers for the Venturi channel confirm the negligible influence of scale effects. The Reynolds number for all flow conditions is above 10^4 , indicating that viscous effects are minimal and the flow remains predominantly turbulent, dominated by inertial forces. The upstream Froude number remains below 1 in all cases, signifying subcritical flow conditions. This aligns with the expected behavior of the Venturi channel, where velocity increases through the constriction but does not transition into supercritical flow. The Weber number for the flow conditions is well above 50, suggesting that surface tension effects are negligible. This ensures that phenomena such as wave formation or capillary effects do not significantly impact the hydraulic performance.

3. Results

3.1. Verification of numerical results

This study evaluates upstream flow velocity (V_1), upstream flow depth (h_1), and discharge coefficient (m) within a Venturi channel through CFD simulations validated against experimental data from Zuikov and Bakunyaeva [1]. The APE error for V_1 , h_1 , and m are 4.12 %, 1.88 %, and 7.36 %, respectively, indicating reasonable agreement between numerical predictions and laboratory measurements as detailed

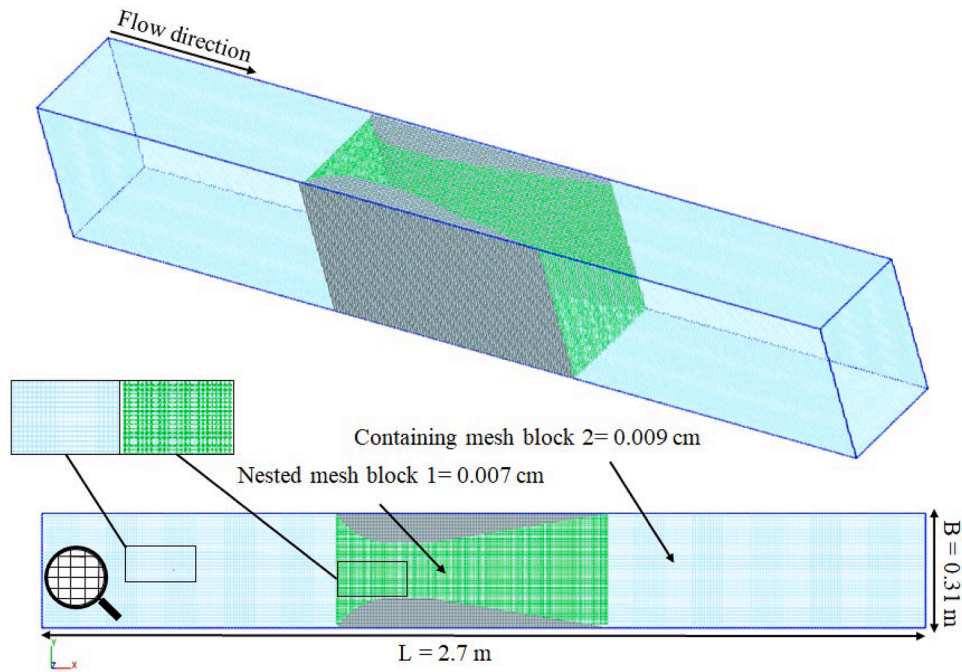


Fig. 4. The meshing of the domain.

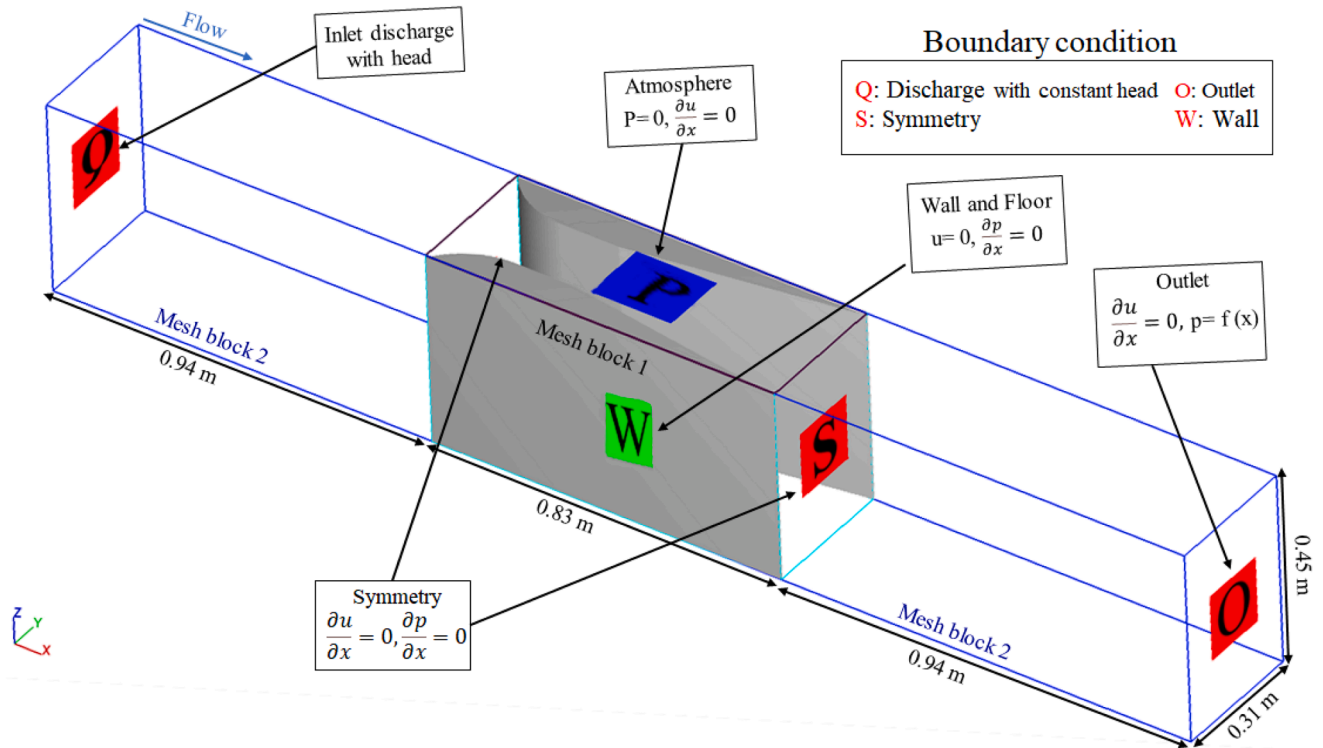


Fig. 5. Boundary conditions in the domain.

in Table 5 and illustrated in Fig. 7. Notably, errors tend to increase at lower flow rates, particularly for the discharge coefficient. This can be attributed to several factors: the RNG $k-\epsilon$ turbulence model assumes fully turbulent flow and constant wall roughness, which may not capture transitional flow regimes and near-wall viscous effects dominant at low discharges. Additionally, idealized geometry and uniform roughness in simulations contrast with slight manufacturing imperfections and variable surface conditions in physical experiments, affecting near-wall

losses. Measurement uncertainties, especially sensor limitations in resolution and calibration at low velocities and depths, also contribute to discrepancies. Moreover, flow regime changes such as unsteadiness, flow separation, and secondary currents become more pronounced at low flow rates, challenging accurate numerical prediction. Finally, numerical discretization errors and mesh sensitivity may have greater impact under these conditions due to weaker velocity gradients and turbulence intensities. Despite these limitations, the model

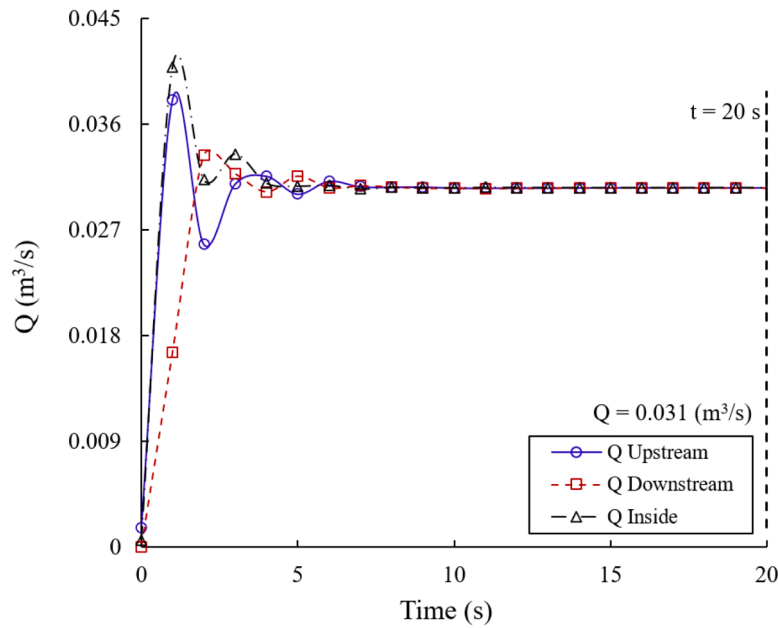


Fig. 6. Solution convergence for inflow and outflow and inside of the constriction discharges $Q = 0.031 \text{ m}^3/\text{s}$ in upstream, downstream and inside the Venturi channel.

Table 5

Comparison between the experimental and numerical results.

Test No.	$Q \text{ (m}^3/\text{s)}$	$V_1 \text{ Exp (m/s)}$	$V_1 \text{ Num (m/s)}$	$h_1 \text{ Exp (m)}$	$h_1 \text{ Num (m)}$	$m \text{ Exp}$	$m \text{ Num}$	$\text{APE}_{V_1} \text{ (\%)}$	$\text{APE}_{h_1} \text{ (\%)}$	$\text{APE}_m \text{ (\%)}$
1	0.031	0.432	0.410	0.228	0.229	0.204	0.190	5.089	0.532	6.775
2	0.026	0.411	0.392	0.208	0.209	0.203	0.190	4.510	0.484	6.571
3	0.023	0.386	0.369	0.186	0.188	0.202	0.188	4.481	1.194	6.793
4	0.018	0.358	0.344	0.162	0.163	0.201	0.187	4.007	0.572	6.795
5	0.014	0.329	0.315	0.136	0.139	0.201	0.186	4.273	2.205	7.623
6	0.010	0.290	0.283	0.107	0.111	0.200	0.184	2.456	3.564	7.800
7	0.006	0.240	0.229	0.074	0.075	0.201	0.184	4.446	1.743	8.366
8	0.001	0.157	0.151	0.030	0.031	0.206	0.189	3.721	4.764	8.163
MEAN								4.12	1.88	7.36

demonstrates robust performance across the tested flow range, supporting its applicability for Venturi channel simulations.

3.2. Discharge depth curve

In Fig. 8, the discharge-depth curve for different models of Venturi channel is shown. As the contraction ratio (b/B) decreases, the flow discharge (Q) also decreases, whereas the flow depth (h) increases. This occurs due to the greater resistance of the fluid in the channel and the reduced flow concentration in the gap. With the increase in the contraction ratio (b/B), the discharge Q increases, but the flow depth h and decrease. In this case, the flow becomes more concentrated, and the fluid velocity increases, resulting in faster discharge. Reducing the contraction ratio is suitable for applications requiring greater flow depth, though it may lead to reduced discharge. Increasing the contraction ratio is appropriate for faster flow discharge and higher flow rates, but it reduces the flow depth. The performance trend lines plotted by the data are the best-fit lines as a function of $Q = kh^a$. Table 6 presents the estimated constants (k , a) for different contraction ratios, showing a strong correlation ($R^2 = 0.99$) between discharge and flow depth for all cases. For the given discharge on average, the flow depth is increased by 17.88 % contraction ratio ($b/B = 0.35$) and for contraction ratio ($b/B = 0.65$) this number is 11.56 %. The numerical results for Zuikov and Bakunyaeva [1] with a contraction ratio ($b/B = 0.50$) were 14.56 %. A contraction ratio of ($b/B = 0.35$) results in an 8 % increase in flow depth compared to the intermediate case ($b/B = 0.50$), due to increased flow

resistance and reduced concentration. Conversely, increasing the contraction ratio ($b/B = 0.65$) results in a 3 % decrease in flow depth compared to the same intermediate case, demonstrating the effect of faster discharge and increased flow concentration. These findings underscore the sensitivity of flow depth to changes in contraction ratio, with lower ratios favoring higher depth and higher ratios reducing it.

3.3. Discharge coefficient

A Venturi channel follows a similar design principle, where the flow rate is a function of the pressure drop across the constriction, the channel's cross-sectional area, and its geometric slope. Zuikov and Bakunyaeva [1] proposed an equation to determine the discharge coefficient (m), allowing designers to evaluate the influence of parameters such as channel width, flow conditions, and slope on flow behavior.

$$m = \frac{Q}{B\sqrt{2gH_1^{1.5}}} \quad (9)$$

Here m is the discharge coefficient, Q is the flow discharge, B is the channel width, g is the gravitational acceleration, and H_1 is the upstream water head (m). The discharge coefficient (m) represents the efficiency of flow through the constriction, incorporating effects of velocity loss, pressure drop, and geometric influence. It is a crucial parameter for determining the discharge through a structure. Eq. (9) can be applied to calculate the discharge coefficient for each Venturi channel with various constriction designs. Fig. 9 presents the variation of m as a function of L/B .

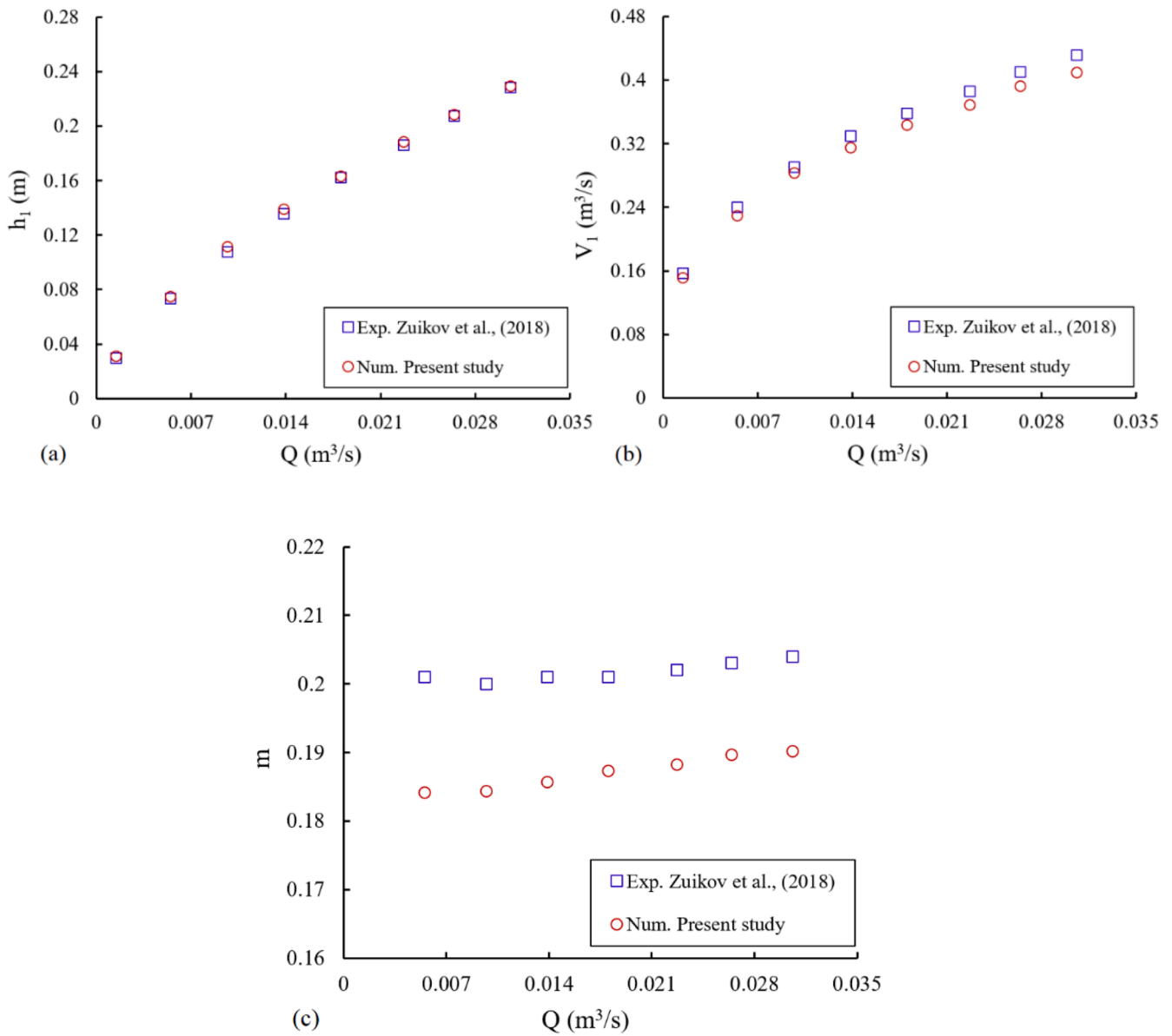


Fig. 7. Comparison of numerical results and experimental data; a) Flow depth. b) Flow velocity c) Discharge coefficient.

H and B/H . It is observed that, for Venturi channel contraction ($b/B = 0.65$), an increase in L/H and B/H increases to m . Conversely, for modified deflections such as contraction ($b/B = 0.35$), the relationship between L/H and B/H and m follows the same trend, with a decrease in m as L/H and B/H increase. Comparing the different shapes, it is evident that the Venturi channel, such as contraction ($b/B = 0.35$) has a lower discharge coefficient than both ($b/B = 0.65$) and ($b/B = 0.50$) for any given discharge or L/H and B/H . On average, the m of ($b/B = 0.35$) is 24.42 % greater than ($b/B = 0.50$) and 18.71 % greater than ($b/B = 0.35$) with 13.62 % as shown in Fig. 9.

3.4. Free surface elevation (F.S.E)

As shown in Fig. 10, the flow profile passing through the three Venturi channel constriction models is presented for four different flowrates $Q = 0.031, 0.018, 0.010$, and $0.001 \text{ m}^3/\text{s}$. The results were obtained from locations upstream of the Venturi channel to the downstream Venturi outlet. Fig. 11 is shown that the flow profiles through the three Venturi channel constriction models, reveal how flow rate affects

the velocity distribution and overall flow behavior. The results, taken from locations just upstream of the Venturi inlet to the outlet downstream, demonstrate a clear relationship between flow rate and the effect of the constriction geometry on the flow. As the flow rate increases, the flow profile at the constriction becomes more pronounced, with higher velocities observed in the constricted section for higher flow rates (e.g., $Q = 0.031 \text{ m}^3/\text{s}$). Conversely, lower flow rates (e.g., $Q = 0.001 \text{ m}^3/\text{s}$) result in a smoother flow through the channel. The geometry of the constriction models significantly influences these velocity variations, with different shapes producing distinct flow profiles. This suggests that the design of the constriction plays a crucial role in determining the flow behavior, which can lead to varying levels of efficiency and potential turbulence.

3.4. Velocity profile and Froude number

Fig. 12 presents a comparison of the longitudinal flow velocity at different contraction ratios in the Venturi channel. As the contraction increases, the upstream flow depth rises along the flow direction within

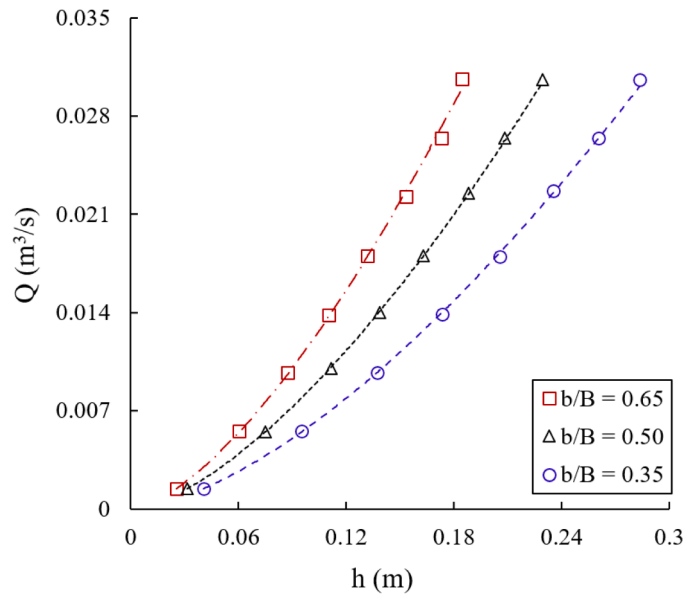


Fig. 8. Discharge depth curves for Venturi channel with different contraction ratio.

Table 6

Estimated k and coefficients and R-squared values for the Venturi channel ($Q = kh^a$).

	Coefficients		R^2	$Q \text{ (m}^3\text{/s)}$
	k	a		
$b/B = 0.65$	0.39	1.52	0.99	$0.39(h)^{1.52}$
$b/B = 0.50$	0.29	1.53	0.99	$0.29(h)^{1.53}$
$b/B = 0.35$	0.21	1.56	0.99	$0.21(h)^{1.56}$

the contraction section. With greater constriction, the curvature of the streamlines in the flow direction intensifies as they approach the contraction. However, a significant variation in flow velocity occurs within the contraction zone, where the velocity reaches its maximum downstream along the channel for all contraction ratios. Additionally, an increase in contraction leads to a higher flow velocity inside the channel, while a decrease in contraction limits the range of velocity variations within the channel and reduces the maximum velocity in the

downstream section.

Fig. 13 shows the velocity profiles on Venturi channel with three constrictions on flow rates $Q = 0.031 \text{ m}^3/\text{s}$. The control sections for velocity measurements along the channel, extracted from upstream, inside the Venturi, and downstream regions, are taken in six points to examine, as shown in Fig. 1b. The velocity values on the Venturi represent the resultant velocity, while the channel velocities correspond to the velocity component in the flow direction. The results indicate that subcritical flow, characterized by lower velocities, transitions to critical flow approximately at the constriction part and then another area of the channel progresses as supercritical flow downstream of the channel. Velocity values increase with higher inflow Froude numbers. The velocity profiles along the Venturi channel were analyzed at six cross-sections $x = -0.15 \text{ m}$, 0 m , 0.12 m , 0.3 m , 0.57 m , 0.83 m , for three constrictions under a constant flow rate of $Q = 0.031 \text{ m}^3/\text{s}$. The results reveal the impact of contraction ratios on flow characteristics and their evolution along the channel. Upstream, such as at $x = -0.15 \text{ m}$, the velocity profiles are nearly symmetric, with maximum velocities occurring at the centerline. The influence of the contraction ratio is minimal in this region as the flow remains relatively undisturbed. At the throat, such as $x = 0$, the contraction effect becomes dominant. Smaller contraction ratios ($b/B = 0.35$) lead to sharper velocity gradients and higher velocity peaks due to the accelerated flow through the narrow section. Conversely, larger ratios ($b/B = 0.65$) exhibit smoother profiles with lower maximum velocities, reflecting less significant acceleration. At downstream, in $x = 0.57 \text{ m}$ and $x = 0.83 \text{ m}$ the velocity profiles begin to flatten and recover symmetry as the flow exits the throat. This recovery process is quicker and more uniform for larger contraction ratios ($b/B = 0.65$), while smaller ratios ($b/B = 0.35$) show lingering velocity gradients due to the higher acceleration experienced at the throat. Overall, the contraction ratio significantly influences the velocity distribution. Smaller ratios generate higher velocities and sharper gradients, while larger ratios result in smoother profiles and faster downstream recovery. These findings highlight the importance of selecting appropriate contraction ratios in Venturi channel designs to balance flow acceleration and energy dissipation.

Fig. 14 provides the flow dynamics within a channel featuring a variable cross-section, emphasizing the distribution of the Froude number and flow depth. The upper section of the figure presents contour plots depicting these parameters across three distinct regions of the channel: (1) the inlet section, where the flow enters the constriction; (2) the gorge section, representing the narrowest segment; and (3) the outlet section, where the channel expands downstream. In the left contour plot,

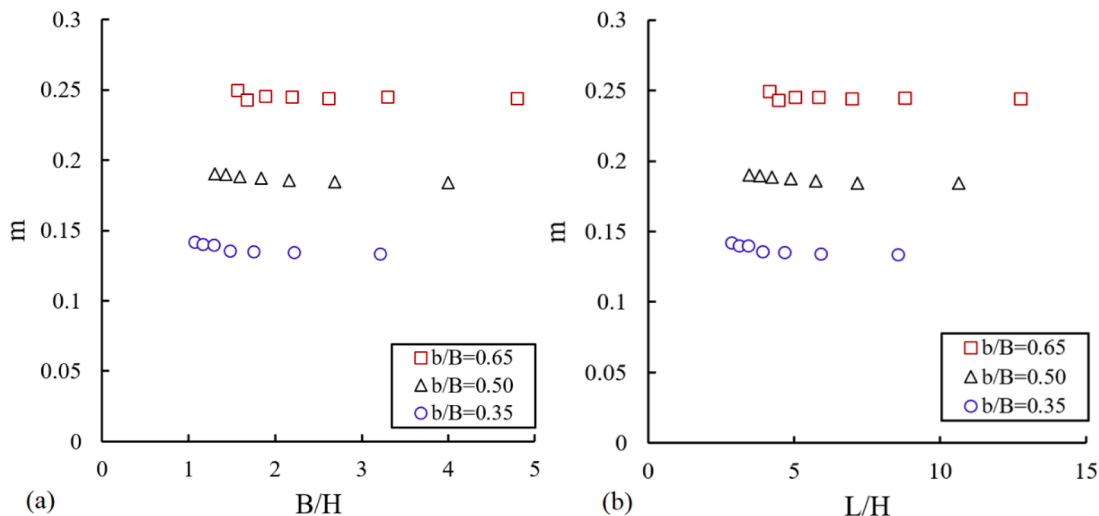


Fig. 9. Discharge coefficient for different b/B ratios; a) Channel width to head height ratio (B/H), b) Venturi length to head height ratio (L/H).

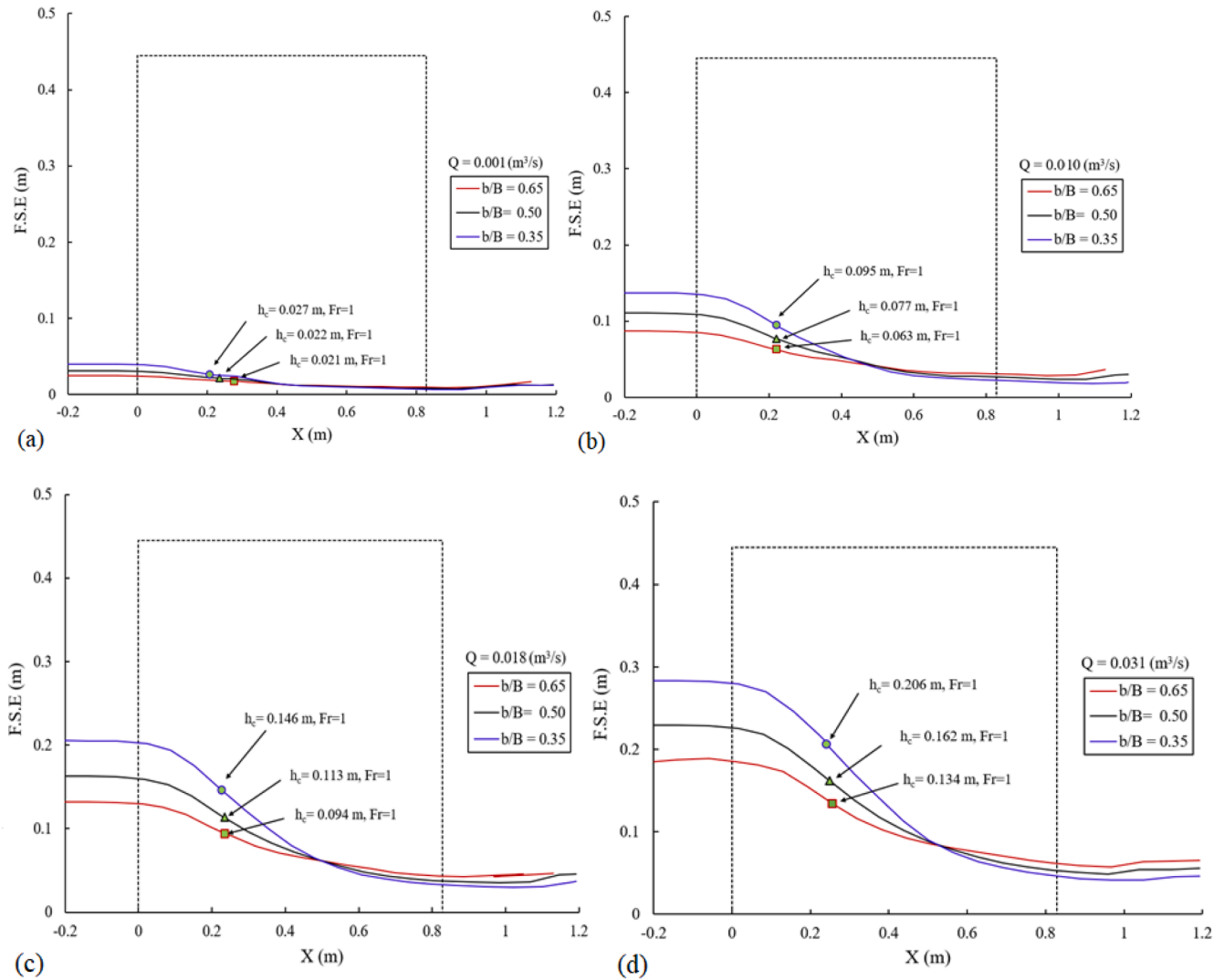


Fig. 10. Flow profile passing through the Venturi model at three different constrictions for varying flow rates.

a) $Q = 0.001 \text{ m}^3/\text{s}$ b) $Q = 0.010 \text{ m}^3/\text{s}$ c) $Q = 0.018 \text{ m}^3/\text{s}$ d) $Q = 0.031 \text{ m}^3/\text{s}$.

the Froude number distribution is visualized, with cooler colors (blue) signifying lower values indicative of subcritical flow, and warmer colors (yellow to red) representing higher values associated with supercritical flow. Notably, the transition from subcritical to supercritical conditions occurs within the gorge section due to flow acceleration. Meanwhile, the right contour plot illustrates the flow depth, which is relatively higher at the inlet, decreases through the gorge as the flow accelerates, and then further decreases in the outlet section due to deceleration. The lower section of the figure contains two-line graphs that provide a comparative analysis of the Froude number and normalized flow depth along the channel for varying contraction ratios (b/B) while maintaining a constant discharge rate ($Q = 0.031 \text{ m}^3/\text{s}$). The left graph depicts the variation of the Froude number along the channel length (x), demonstrating that a decrease in the contraction ratio results in more pronounced flow acceleration in the gorge section, leading to a higher peak in the Froude number. Conversely, the right graph illustrates the normalized flow depth (h/E) along the channel, indicating that the depth decreases through the narrow gorge before gradually recovering in the outlet region. The primary distinction between these two plots lies in their focus: the left graph highlights the effects of flow acceleration and the transition to supercritical conditions, whereas the right graph emphasizes the corresponding variations in flow depth. Collectively, these visualizations effectively depict the intricate relationship between channel

geometry, flow velocity, and depth, demonstrating how channel narrowing influences both the Froude number and flow depth dynamics.

3.6. Turbulence characteristics

Turbulence characteristics are important for the Venturi channel. Estimating and analyzing turbulent kinetic energy TKE in Venturi channel is crucial as it influences in contraction site. Velocity and flow depth can affect the channel. In this study, flow rates: $Q = 0.031 \text{ m}^3/\text{s}$. The control sections for TKE along the channel, extracted from upstream, inside the Venturi, and downstream regions in sum six point taken to examined Fig. 1b Turbulent kinetic energy is defined as follows Ahmadi et al., (2021):

$$TKE = \frac{1}{2} (u_x'^2 + u_y'^2 + u_z'^2) \quad (10)$$

where u_x , u_y , and u_z are fluctuating velocities in the x , y , and z directions, respectively. Turbulent dissipation (TD) is one of the flow variables that determines the state of turbulence and helps better understand the multiple concepts related to the effects of turbulence parameters. Its importance has been widely considered in several texts Lumley & Blossey, [22]. The TD is defined as follows:

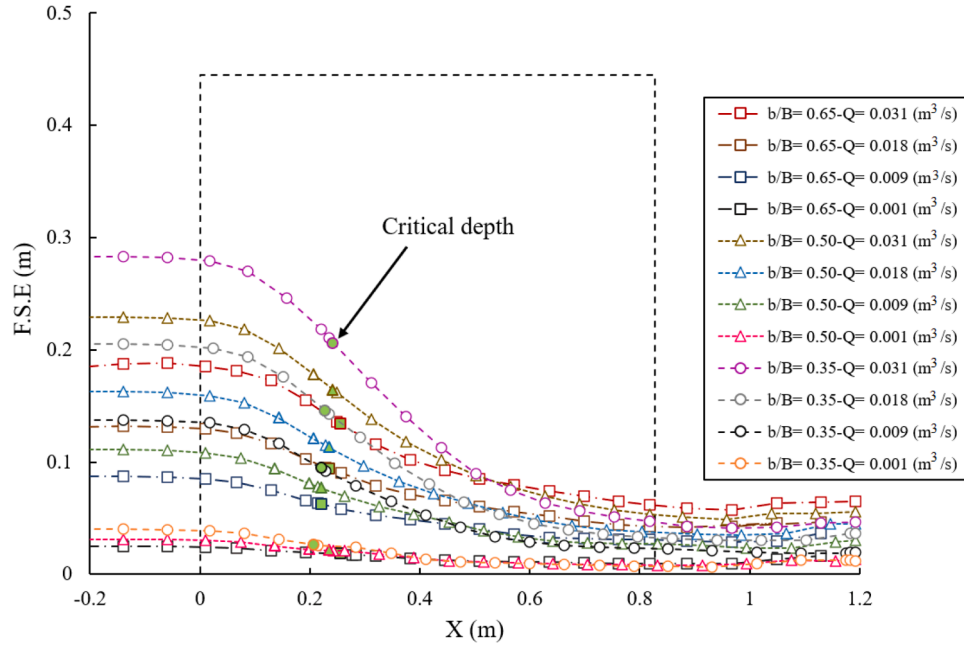


Fig. 11. Flow pattern passing through the three Venturi constriction models.

$$TD = v \left(\frac{\partial u'_x}{\partial x} + \frac{\partial u'_y}{\partial x} \right) \quad (11)$$

where v is the kinematic turbulent viscosity, there are several low-turbulence flow corridors with different hydraulic conditions. The values of turbulence intensity (TI) in the Mean Surface Curvature Weighting (MSCW) are examined. TI is defined according to Eq (12) and represents the ratio of the standard deviation of velocity fluctuations to the average velocity V [23].

$$TI(\%) = \frac{\sqrt{\frac{1}{3}(\overline{u'^2} + \overline{v'^2} + \overline{w'^2})}}{\overline{V}} \times 100 \quad (12)$$

where u' , v' , and w' represent velocity fluctuations in the x , y , and z directions as we go through.

Fig. 15 illustrates the normalized Turbulent Kinetic Energy TKE distribution (TKE/TKE_{max}) along the normalized vertical axis (z/z_{max}) for three different constriction ratios ($b/B = 0.35, 0.50$ and 0.65) in a Venturi channel. These measurements were taken at six streamwise positions (x), ranging from upstream of the constriction ($x = -0.15$ m) to downstream ($x = 0.83$ m), under a constant flow rate of $Q = 0.031$ m³/s. The analysis reveals how constriction ratios and streamwise positions influence the turbulence characteristics within the flow. Upstream of the constriction ($x = -0.15$ m), the TKE values are relatively low, increasing gradually toward the channel center for all constriction ratios. Narrower constrictions ($b/B = 0.35$) exhibit slightly higher turbulence intensity, suggesting that the effect of channel narrowing begins upstream of the constriction. This trend is amplified at the start of the constriction ($x = 0$ m), where TKE increases significantly, especially near the centerline. The turbulence is more intense for narrower constrictions ($b/B = 0.35$), indicating that greater channel narrowing amplifies the energy of turbulent eddies as the flow accelerates into the constriction. Inside the constriction ($x = 0.12$ m), the TKE distribution becomes more concentrated along the centerline, with lower turbulence levels near the channel walls. The peak turbulence intensity is more pronounced for narrower constrictions, reflecting the stronger velocity gradients and enhanced mixing caused by the reduced cross-sectional area. As the flow exits the constriction ($x = 0.30$ m), TKE begins to dissipate, although the turbulence persists for a longer distance in narrower constrictions. This slower dissipation rate highlights the lasting effect of intense turbulence

generated by the sharper constriction. Further downstream ($x = 0.57$ m), turbulence continues to decay, and the TKE profile becomes more uniform, indicating a gradual stabilization of the flow. Narrower constrictions still exhibit slightly higher turbulence levels compared to wider ones. By the time the flow reaches ($x = 0.83$ m), TKE values have significantly reduced, approaching equilibrium. However, the differences between constriction ratios are still apparent, with ($b/B = 0.35$) maintaining slightly higher turbulence levels than the other configurations.

Fig. 16 illustrates the transverse distribution of TKE along the Venturi channel. Near the upstream section of the channel, TKE values are minimal and negligible. As the flow progresses into the channel and passes through the constrictions, the highest TKE values are observed in all three constrictions, particularly near the bottom and sides of the channel. Moving away from the bottom and sides, towards the center of the channel's width and height, and approaching the flow surface, TKE values exhibit a sharp decline. As the constrictions of the channel increases, the TKE within the channel increases, especially along the bottom and side walls. No significant difference in TKE distribution is observed between the constrictions for the designs with $b/B = 0.50$ and $b/B = 0.65$. However, the design with $b/B = 0.65$ shows a larger region of low TKE compared to the $b/B = 0.50$ design.

Fig. 17 shows the turbulence intensity TI, a key parameter used to describe the level of turbulence in a fluid flow. It provides a dimensionless measure of the velocity fluctuations within a flow relative to the mean velocity. TI is commonly used in engineering and fluid dynamics to assess flow characteristics, energy losses, and mixing efficiency. In Fig. 16 at ($x = -0.15$ m) this position, the turbulence intensity TI is generally low across the vertical profile. However, narrower constrictions ($b/B = 0.35$) show slightly higher TI values compared to wider constrictions, particularly near the centerline. This indicates that the influence of the constriction extends slightly upstream, initiating turbulence intensity even before the flow reaches the constriction. In ($x = 0$ m) as the flow approaches the constriction, TI increases sharply near the centerline, particularly for narrower constrictions ($b/B = 0.35$). This sharp increase is caused by the accelerated flow and the associated velocity gradients near the constriction. Wider constrictions ($b/B = 0.65$) exhibit relatively moderate TI values, reflecting smoother flow transitions. Inside the Constriction ($x = 0.12$ m) TI peaks in this region, with the highest values observed near the centerline for the narrowest

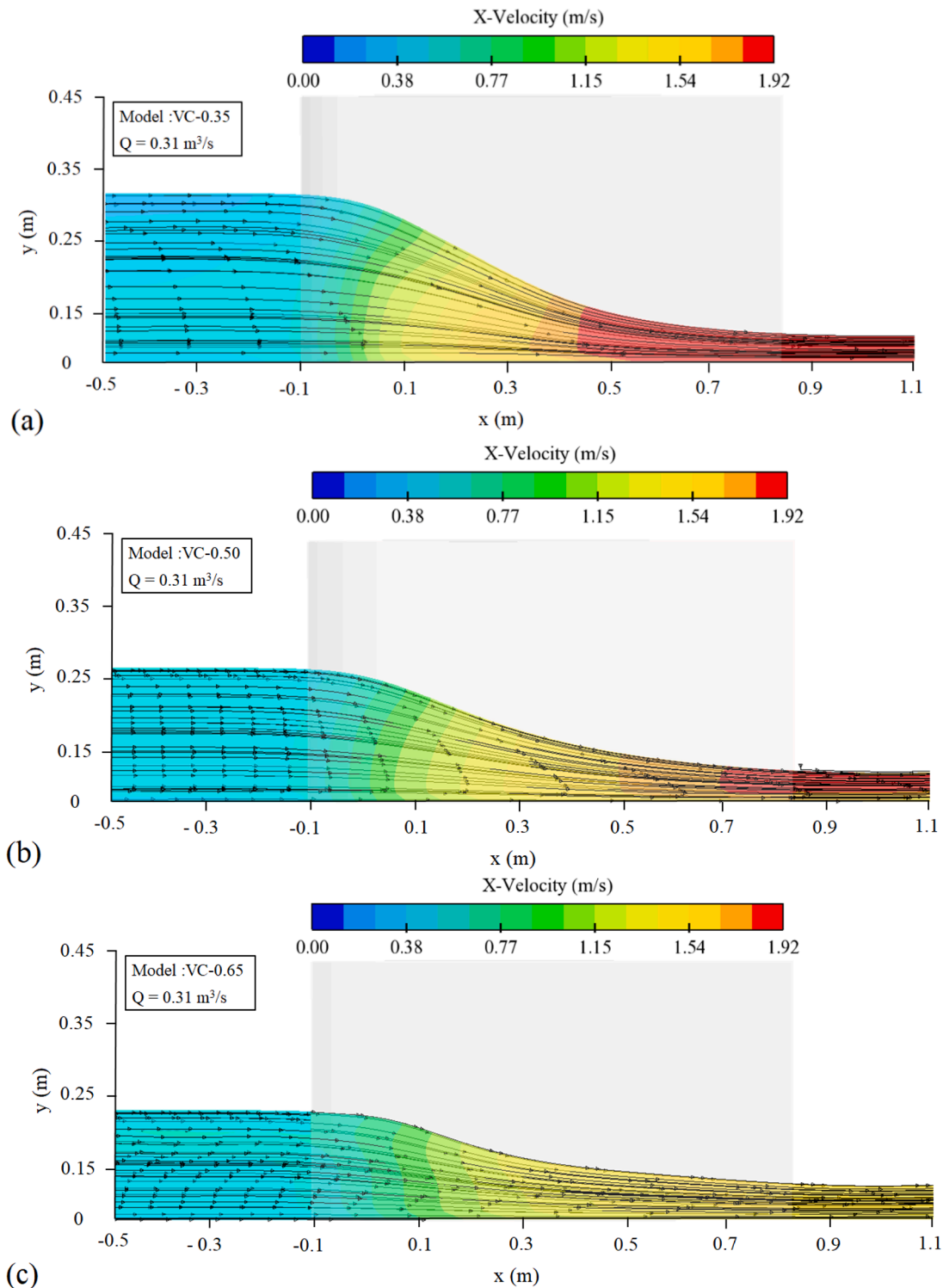


Fig. 12. Longitudinal velocity flow contours through a Venturi channel at varying constriction. a: $b/B = 0.35$ b: $b/B = 0.50$ c: $b/B = 0.65$.

constriction ($b/B = 0.35$). This highlights the strong turbulence generation due to the significant reduction in the cross-sectional area. TI values near the walls are considerably lower, indicating that the turbulence is concentrated in the central region of the flow. Exit of the Constriction ($x = 0.30$ m) Upon exiting the constriction, TI begins to decrease, but narrower constrictions ($b/B = 0.35$) maintain higher TI levels compared to wider ones. This indicates that the turbulence effects persist for a longer distance downstream for narrower constrictions,

delaying flow stabilization. Further downstream ($x = 0.57$ m) TI continues to decay at this position, and the profiles become more uniform across the channel height. Wider constrictions ($b/B = 0.65$) exhibit significantly lower TI values, reflecting faster stabilization and reduced turbulence intensity. Conversely, narrower constrictions ($b/B = 0.35$) still maintain higher TI levels, demonstrating a longer recovery length. Far Downstream ($x = 0.83$ m) at this location, TI values are significantly reduced, indicating that the flow has largely stabilized. The differences

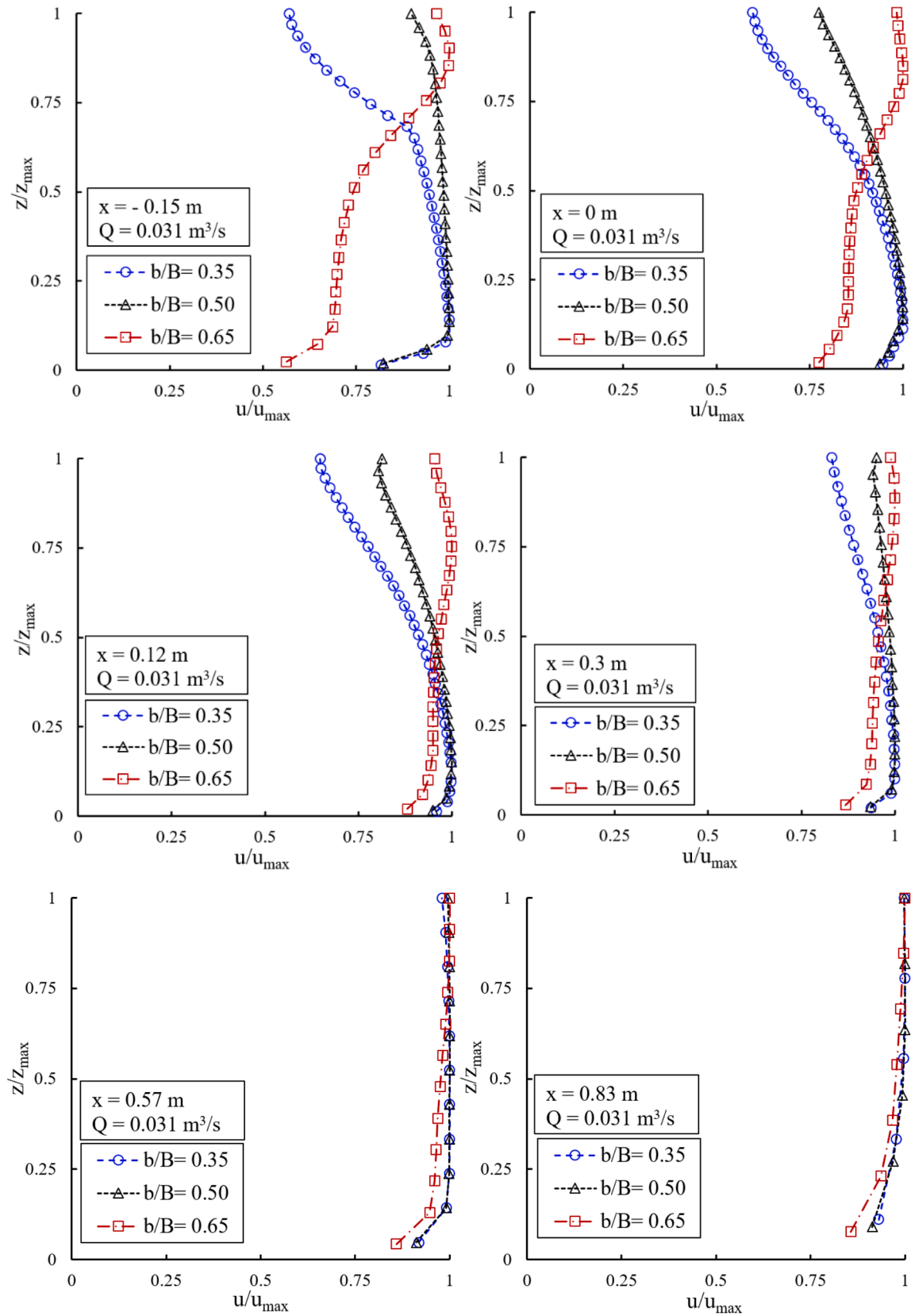


Fig. 13. Dimensionless velocity profile for three constrictions for $Q = 0.031 \text{ m}^3/\text{s}$.

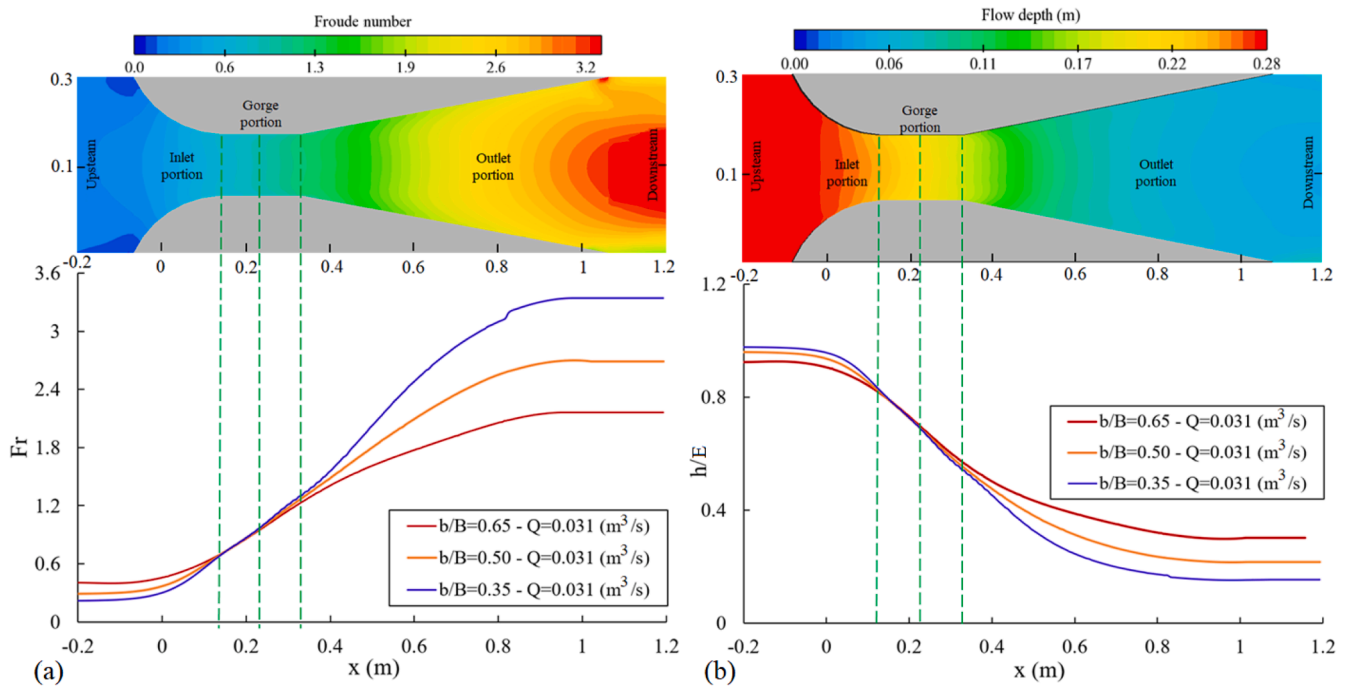


Fig. 14. Diagram of Froude number and flow depth; a: Froude number b: Flow depth.

between the constriction ratios become less pronounced, but narrower constrictions ($b/B = 0.35$) still show slightly higher TI levels, reflecting the lingering effects of higher turbulence generation upstream.

Fig. 18 presents the transverse distribution of TI from the upstream to the downstream section of the Venturi channel. Upstream, where the flow initially enters the channel, TI values are higher along the sides and bottom of the channel compared to the flow surface in all three constrictions. As the flow progresses through the channel, TI values increase, reaching their peak at the downstream end. With increasing constrictions, TI within the channel increases, particularly along the bottom and side walls. The variation in TI distribution along the channel, from upstream to downstream, is more pronounced in the $b/B = 0.35$ design compared to the $b/B = 0.50$ and $b/B = 0.65$ designs. However, the design with $b/B = 0.65$ exhibits a larger region of low TI relative to the $b/B = 0.50$ design.

In Fig. 19 upstream of the Constriction ($x = -0.15$ m) At this position, TD values are relatively low and exhibit a slight increase toward the centerline. This indicates minimal turbulence dissipation activity before the constriction. However, for the narrower constriction ratio ($b/B = 0.35$), TD values are slightly higher, suggesting that the influence of the constriction extends upstream, initiating energy dissipation even before the flow enters the constriction. At the Start of the Constriction ($x = 0$ m) as the flow approaches the constriction, TD increases sharply, especially near the centerline. Narrower constrictions ($b/B = 0.35$) show higher TD values compared to wider constrictions due to sharper velocity gradients and increased shear effects. This sharp increase reflects the turbulence intensification caused by the flow acceleration near the constriction. Inside the Constriction ($x = 0.12$ m) TD values reach their peak near the centerline in this region, with significantly lower values near the walls. The reduced cross-sectional area in narrower constrictions ($b/B = 0.35$) amplifies turbulence dissipation rates, leading to higher TD peaks compared to wider constrictions. This region represents the maximum energy dissipation due to turbulence. Exit of the Constriction ($x = 0.30$ m) After the flow exits the constriction, TD values decrease but remain concentrated near the centerline. Narrower constrictions maintain higher TD levels, indicating that the turbulence effects persist longer. The high dissipation rates at this position reflect the delayed flow stabilization for sharper constrictions. Further downstream

($x = 0.57$ m) TD continues to decay as the flow progresses downstream, and the profiles become more uniform across the vertical axis ($z/z_{\max}$). Narrower constrictions ($b/B = 0.35$) still exhibit higher TD values, suggesting a longer recovery length. In contrast, wider constrictions ($b/B = 0.65$) enable faster flow stabilization, with lower TD values observed at this position. Far downstream ($x = 0.83$ m), TD values are significantly reduced at this position indicating that the flow has largely stabilized. The differences between the constriction ratios diminish, but narrower constrictions ($b/B = 0.35$) still retain slightly higher TD levels, highlighting the lasting impact of intense turbulence dissipation generated by the sharp constriction.

Fig. 20 illustrates the transverse distribution of TD along the Venturi channel. In all three constriction scenarios, TD values are minimal and negligible in the upstream section of the channel. As the flow advances through the channel and passes through the constrictions, the highest TD values are observed, particularly near the bottom and sides of the channel, where they reach their peak at the downstream end. Moving towards the center of the channel's width and height, and approaching the flow surface, TD values exhibit a sharp decline. In the designs with a b/B ratio of 0.35, as the constriction of the channel increases, TD within the channel also increases, particularly along the bottom and side walls. However, when the constriction decreases, a region of low TD develops within the channel.

3.7. Energy dissipation rate in Venturi channel

This study examines energy dissipation in a Venturi channel for three different constriction ratios: $b/B = 0.35$, $b/B = 0.50$, and $b/B = 0.65$. Analyzing these ratios provides valuable insights into how varying the degree of constriction affects energy dissipation and helps in determining the most efficient design. The key parameters considered are the energy values E_1 and E_2 at different locations, as well as the energy dissipation ratio ($\Delta E/E_1$) and the residual energy ratios (E_2/E_1). These ratios quantify the amount of energy loss relative to the initial energy and indicate the efficiency of energy conversion as the fluid flows through the channel, as shown in Fig. 21, where these quantities are plotted against the flow rate per unit width (q). For the lower constriction ($b/B = 0.65$), the energy dissipation ratio values are

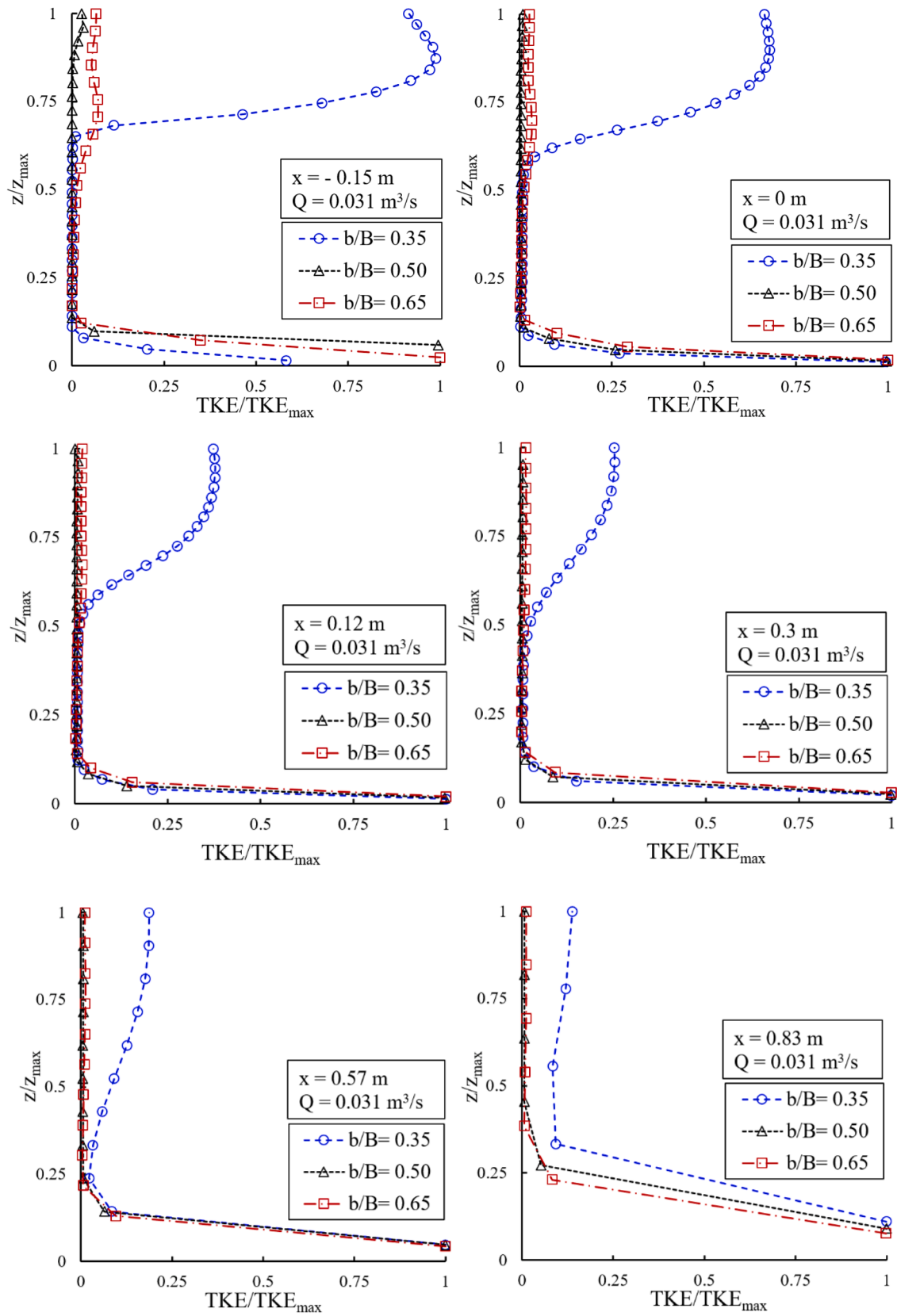


Fig. 15. TKE distribution from different constrictions $Q = 0.031 \text{ m}^3/\text{s}$.

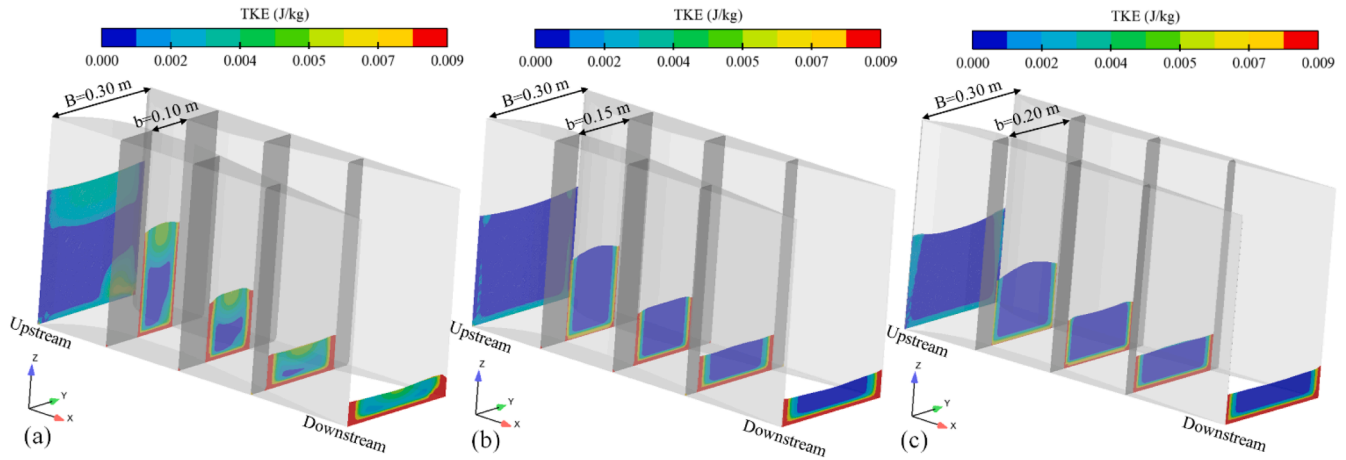


Fig. 16. Cross-sectional distribution of TKE in three different channel constrictions ($Q = 0.031 \text{ m}^3/\text{s}$).
a: $b/B = 0.35$ b: $b/B = 0.50$ c: $b/B = 0.65$.

consistently the lowest across all flow rates, suggesting minimal energy dissipation compared to the other cases. Additionally, the ratios E_2/E_1 remains closer to 1, indicating a more efficient conversion between pressure energy and kinetic energy with minimal losses. This trend persists across all flow rates, demonstrating that lower constriction results in smoother flow, reduced turbulence, and lower energy dissipation. As the constriction increases and the constriction ratio of $b/B = 0.35$, energy dissipation increases. This indicates substantial energy loss due to the intense acceleration of the fluid through the narrowest passage. The increased energy dissipation is attributed to higher velocity gradients, the onset of turbulence, and flow separation at the throat of the Venturi channel. Although the system maintains reasonable efficiency, the overall energy dissipation is noticeably greater than in the least constricted scenario. The extreme velocity gradients lead to heightened turbulence and pronounced flow separation, as confirmed by the highest E_2/E_1 values, signifying inefficient energy conversion. The severe constriction induces chaotic flow patterns, increasing frictional losses and significantly reducing overall efficiency. From this analysis, it is evident that greater constriction leads to increased energy dissipation due to enhanced turbulence and flow instability. Therefore, the least constricted case ($b/B = 0.65$) is the most efficient design, as it minimizes energy dissipation while maintaining stable and streamlined flow. Conversely, the highest constriction ($b/B = 0.35$) should be avoided when energy efficiency is a priority, as it results in the greatest energy dissipation.

4. Discussion

This study provides a detailed analysis of how different constriction ratios in Venturi channels influence hydraulic behavior, turbulence, energy dissipation, and flow stability. The results demonstrate that constriction geometry plays a crucial role in determining flow characteristics, particularly in terms of flow depth, velocity profiles, and energy distribution. The impact of the constriction ratio on flow depth is evident, as narrower configurations ($b/B = 0.35$) result in greater flow depth, but also induce higher velocities and greater energy losses. In contrast, wider constrictions ($b/B = 0.65$) reduce flow depth while promoting a more stable and lower-energy flow. This finding is particularly significant for improving hydraulic system efficiency in various applications. The discharge coefficient (m) is directly influenced by the length-to-height (L/H) and width-to-height (B/H) ratios of the channel. The results indicate that a constriction ratio of $b/B = 0.35$ increases the discharge coefficient by 24.42 % compared to $b/B = 0.50$ and 18.71 % compared to $b/B = 0.65$. However, this also leads to higher turbulence and energy loss, highlighting the need to select an optimal constriction

ratio based on specific application requirements.

For discharge depth curve demonstrating the effect of faster discharge and increased flow concentration. These findings underscore the sensitivity of flow depth to changes in contraction ratio, with lower ratios favoring higher depth and higher ratios reducing it. The discharge coefficient (m) increases with L/H and B/H for a Venturi channel with a contraction ratio of ($b/B = 0.65$), while it decreases for ($b/B = 0.35$). On average, the discharge coefficient for ($b/B = 0.35$) is 24.42 % higher than ($b/B = 0.50$) and 18.71 % higher than ($b/B = 0.65$). Understanding the relationship between flow rate and constriction geometry is essential for optimizing Venturi channel designs. Tailoring the geometry to specific flow conditions helps improve flow efficiency, reduce losses, and minimize issues such as flow separation. Therefore, carefully considering both flow rate and constriction design is necessary to achieve optimal performance in practical applications.

Detailed velocity profiles measured at six cross-sectional locations along the Venturi channel ($x = -0.15 \text{ m}, 0 \text{ m}, 0.12 \text{ m}, 0.3 \text{ m}, 0.57 \text{ m}, 0.83 \text{ m}$) under a constant flow rate ($Q = 0.031 \text{ m}^3/\text{s}$) reveal the nuanced influence of constriction ratios on flow development. Upstream sections exhibit nearly symmetric velocity distributions with minimal influence from the contraction geometry. At the constriction throat, narrower ratios ($b/B = 0.35$) produce pronounced velocity gradients and peak velocities due to flow acceleration within the narrow segment, whereas wider ratios ($b/B = 0.65$) yield smoother velocity profiles with reduced peaks. Downstream recovery of velocity profiles is more rapid and uniform for larger constrictions, while narrower channels maintain steeper velocity gradients over extended distances. The Froude number distributions confirm a flow regime transition from subcritical upstream to supercritical within the throat, more accentuated at lower contraction ratios. Correspondingly, normalized flow depth decreases markedly through the throat and gradually recovers downstream, with deeper upstream flow consistently observed for all geometries.

Turbulent kinetic energy (TKE) analysis reveals narrower constrictions ($b/B = 0.35$) generate higher TKE values and prolong the recovery length of the flow, resulting in enhanced turbulence intensity and mixing. Wider constrictions ($b/B = 0.65$) produce lower peak TKE values and allow for faster dissipation of turbulence downstream. The flow development demonstrates that turbulence begins upstream of the constriction, peaks inside the constriction, and gradually dissipates downstream, with the recovery length strongly influenced by the constriction ratio. These findings provide valuable insights for optimizing Venturi channel designs based on specific engineering requirements, such as enhancing mixing or reducing turbulence intensity. Turbulence intensity (TI) and turbulence dissipation (TD) distributions follow similar trends; smaller constrictions enhance turbulence

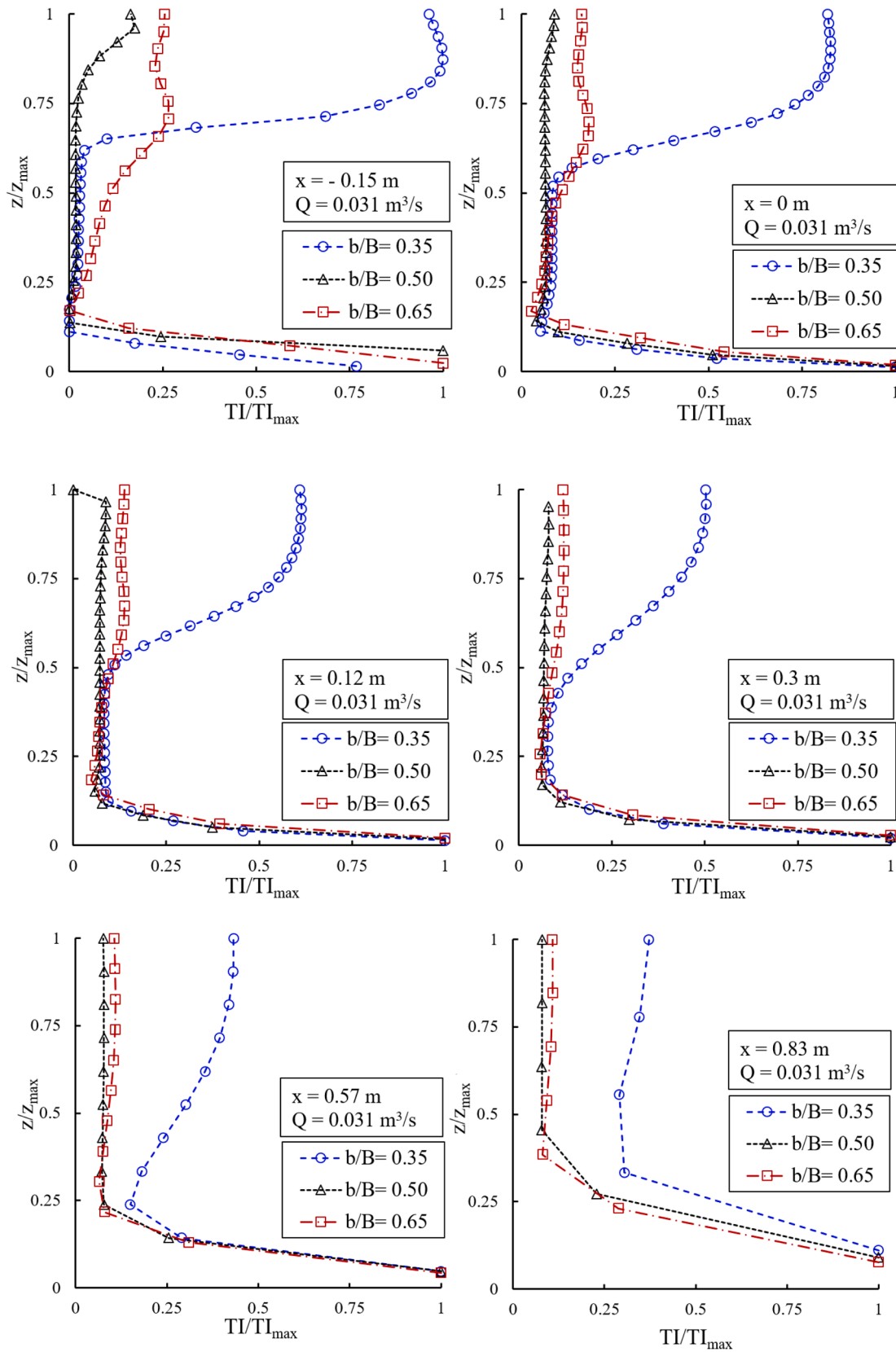


Fig. 17. TI distribution from different constrictions $Q = 0.031 \text{ m}^3/\text{s}$.

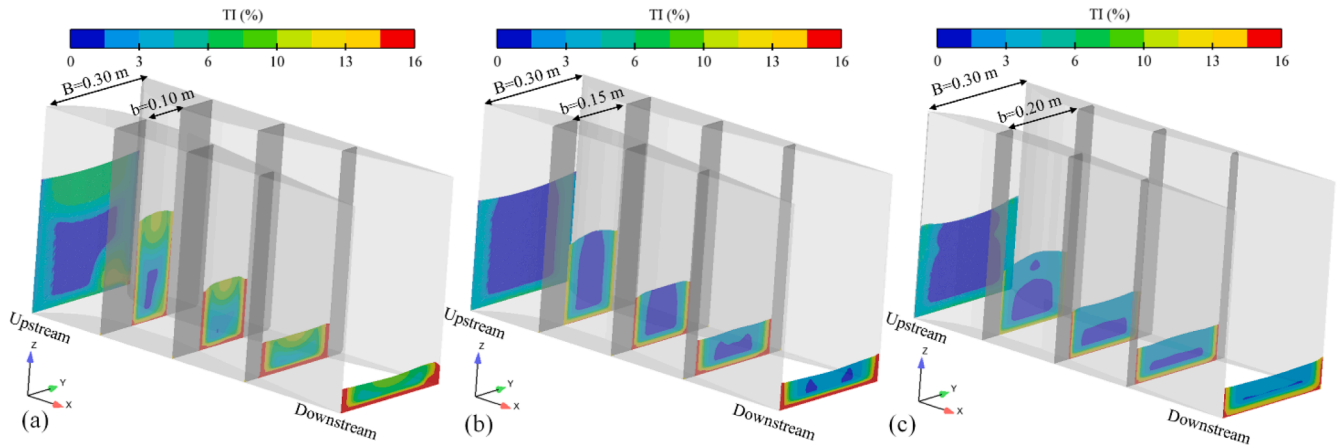


Fig. 18. Cross-sectional distribution of TI in three different channel constrictions ($Q = 0.031 \text{ m}^3/\text{s}$).
a: $b/B = 0.35$ b: $b/B = 0.50$ c: $b/B = 0.65$.

generation and delay flow stabilization, while larger constrictions support smoother flow transitions and reduced turbulence persistence.

Energy dissipation analysis highlights the significant influence of constriction ratios on flow efficiency within the Venturi channel. The configuration with a higher constriction ratio ($b/B = 0.65$) consistently demonstrates superior energy efficiency, characterized by lower energy losses and more effective conversion of pressure energy into kinetic energy. The relatively smooth flow regime at this ratio is associated with reduced turbulence intensity and minimal frictional dissipation, making it optimal for applications prioritizing energy conservation. In contrast, reducing the constriction ratio to $b/B = 0.35$ results in a marked increase in energy dissipation. This escalation is primarily driven by intensified fluid acceleration within the narrowest section, generating steep velocity gradients, heightened turbulence production, and flow separation phenomena. These factors collectively amplify frictional losses and degrade the overall energy conversion efficiency. Furthermore, excessive constriction fosters complex and chaotic flow patterns that impair hydraulic performance and system stability. The consistently elevated residual energy ratios (E_2/E_1) observed in wider constrictions reaffirm their advantage in maintaining energy conversion effectiveness. These findings emphasize the critical need to balance the degree of constriction with flow stability to optimize Venturi channel performance. Accordingly, designs favoring a moderate constriction ratio, such as $b/B = 0.65$, are recommended to achieve minimal energy dissipation while preserving stable, streamlined flow conditions. Conversely, highly constricted geometries, despite their potential benefits in specific applications, should be avoided where energy efficiency is a key objective.

Collectively, these quantitative and qualitative observations highlight the critical balance required in Venturi channel design. While narrower constrictions enhance discharge coefficients and turbulence, beneficial for mixing-intensive applications, they also impose increased energy penalties and prolonged flow recovery. Conversely, wider constrictions favor energy efficiency and flow stability, suited to applications where minimizing losses is paramount. These findings support the strategic selection of constriction ratios aligned with targeted engineering objectives.

4.1. Insights from the numerical analysis and their relevance

The computational results demonstrate that the constriction ratio shapes flow behavior within the Venturi channel. Narrower constrictions (e.g., $b/B = 0.35$) exhibit sharper velocity gradients and higher turbulence intensity, particularly at the constriction throat, compared to wider configurations ($b/B = 0.65$). This behavior highlights the importance of selecting an appropriate constriction ratio based on specific application requirements.

The analysis of energy dissipation indicates that higher constriction ratios, such as $b/B = 0.35$, generate increased turbulence and energy dissipation at lower flow rates due to pronounced velocity gradients and flow separation. However, at higher flow rates, these constrictions exhibit a slight improvement in efficiency as energy dissipation decreases. In contrast, lower constriction ratios, such as $b/B = 0.65$, consistently result in lower energy losses across all flow conditions but are less effective in optimizing flow efficiency at elevated discharge rates.

These findings align with previous research emphasizing the trade-off between energy efficiency and flow stabilization in hydraulic systems Zulkov & Suehtina, [24], the numerical analysis of TKE and turbulent dissipation TD, further underscores the impact of channel narrowing. Constrictions with ($b/B = 0.35$) amplify turbulence near the centerline and prolong the recovery length downstream, creating conditions favorable for enhanced mixing. Broader configurations ($b/B = 0.65$) promote smoother flow transitions and faster stabilization, making them suitable for applications prioritizing uniform flow patterns.

4.2. Limitations

Despite the comprehensive nature of the numerical simulations presented in this study, several inherent limitations must be acknowledged, particularly regarding the generalization of results to real-world applications. Firstly, the study was conducted using a single, fixed geometry with constant channel length, width, and slope. This simplification limits the applicability of the findings to scenarios involving variable slopes or geometries, which are common in natural and industrial settings. Variations in slope and channel dimensions can significantly influence flow recovery length, turbulence intensity, velocity profiles, and energy dissipation, but these effects were not explicitly investigated. Secondly, the simulations focus exclusively on steady-state, single-phase water flow treated as a Newtonian fluid. The presence of sediment transport, which is prevalent in many environmental and engineering contexts (e.g., open channels, wastewater systems), introduces complexities such as altered effective channel roughness and flow stability, which were not modeled here. Similarly, non-Newtonian fluids, characterized by shear-dependent viscosity, would exhibit fundamentally different flow behavior and turbulence characteristics that require separate consideration. While scale effects were briefly discussed in Section 2.5, it is important to note that the CFD simulations rely on turbulence closure models. The RNG $k-\epsilon$ turbulence model employed in this study is well-established for many turbulent flow regimes but has known limitations in accurately capturing transitional flows or strong flow separations that might occur under different flow rates or geometries beyond those studied. The accuracy of

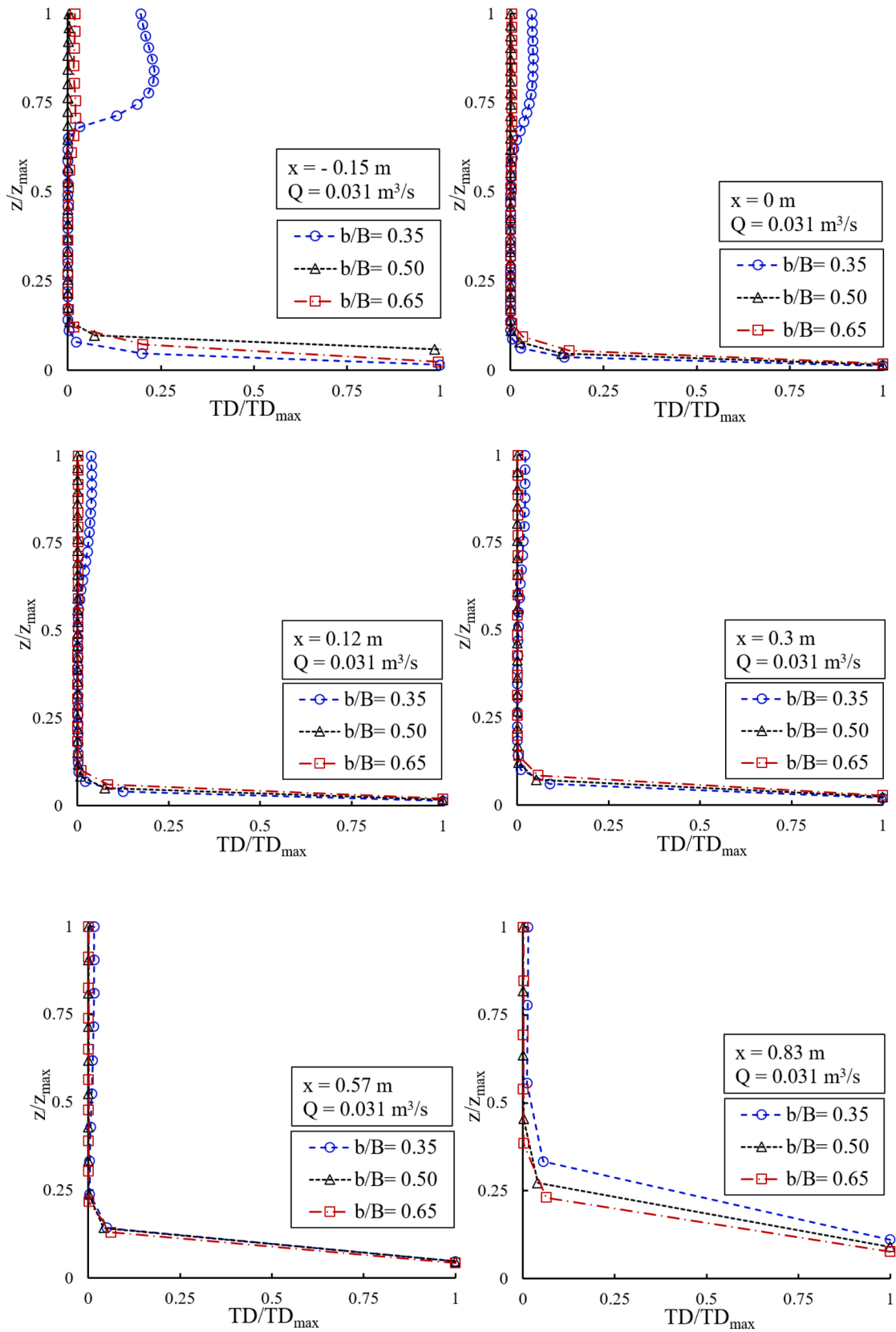


Fig. 19. TD distribution from different constrictions $Q = 0.031 \text{ m}^3/\text{s}$.

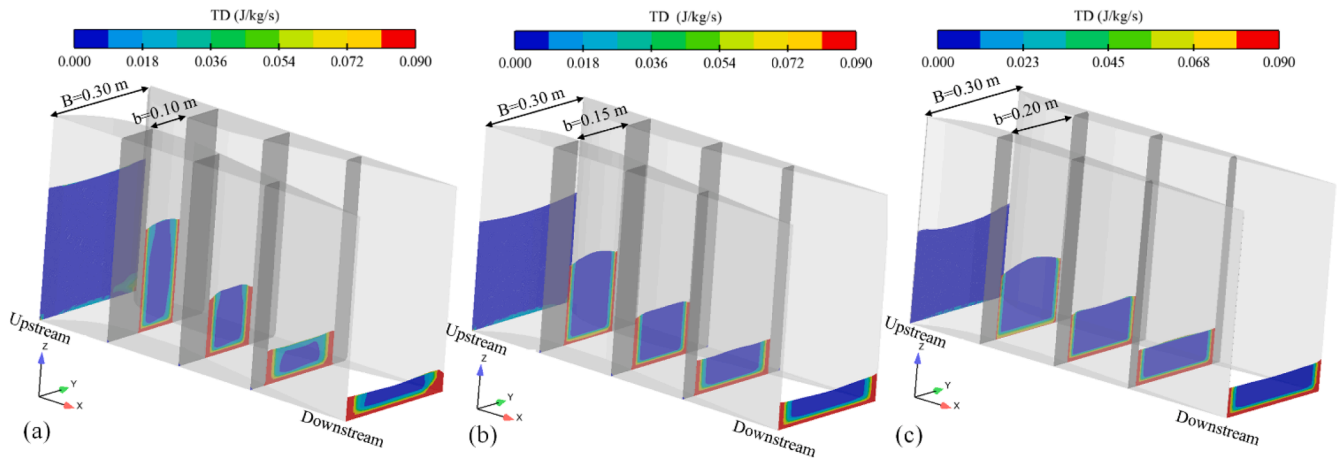


Fig. 20. Cross-sectional distribution of TD in three different channel constrictions ($Q = 0.031 \text{ m}^3/\text{s}$).
a: $b/B = 0.35$ b: $b/B = 0.50$ c: $b/B = 0.65$.

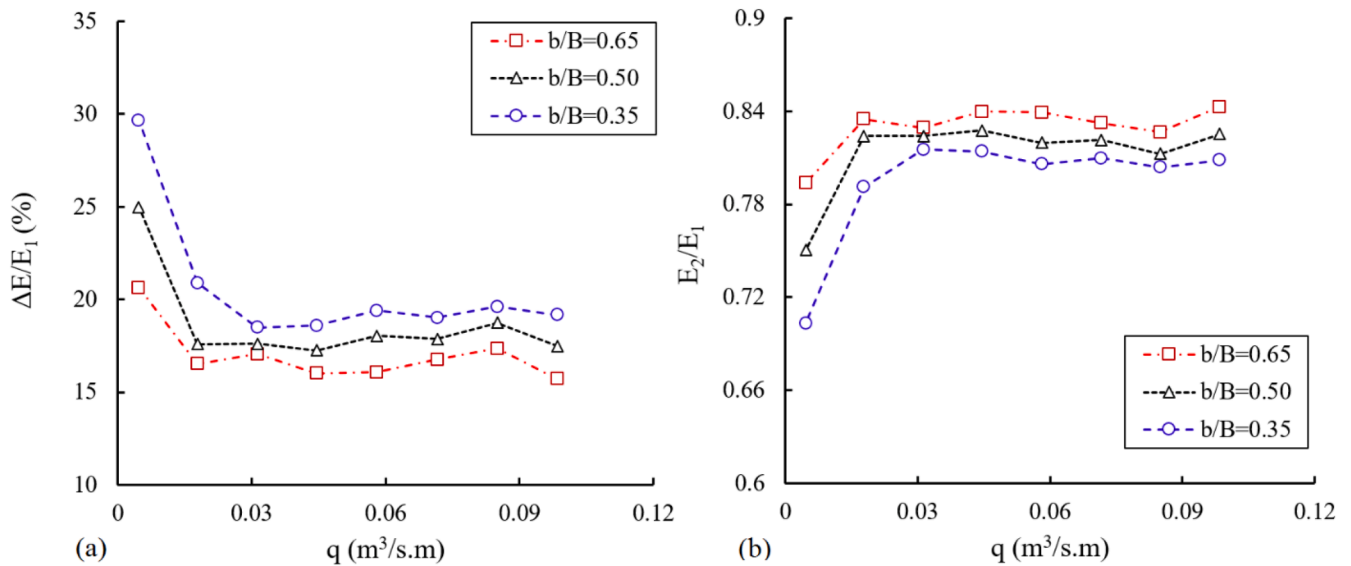


Fig. 21. Energy dissipation and downstream energy versus q .
a: Energy dissipation b: Residual energy.

turbulence models is also scale-dependent; thus, extrapolating results from model scale to prototype scale must be done cautiously. Future work would benefit from incorporating alternative turbulence models, such as SST $k-\omega$ or Large Eddy Simulation (LES), and validating simulations across a wider range of Reynolds numbers. Finally, the lack of field validation limits the direct transferability of the numerical findings. Although good agreement was observed between the simulations and prior experimental data [1], real-world factors such as material heterogeneity, environmental conditions, and operational variability may affect hydraulic performance and should be addressed in future research.

4.3. Recommendations for future research

Future research should aim to address the limitations identified in this study to further enhance the understanding and applicability of Venturi channel systems. Field-scale experiments conducted under a variety of environmental and operational conditions would be particularly valuable for validating the numerical findings and improving their real-world relevance. In addition, extending the analysis to include transient (unsteady) flow conditions, sediment transport, and debris

accumulation would provide a more comprehensive assessment of channel performance in practical scenarios. Investigating variable geometries (e.g., different slopes, widths, and constriction ratios) could lead to optimized designs tailored to specific applications such as wastewater treatment, irrigation networks, and hydropower facilities. Hybrid Venturi configurations that combine multiple constriction ratios may also deliver improved hydraulic performance across a wider range of flow regimes. Emerging computational approaches offer further opportunities for advancement. The integration of artificial intelligence (AI) and machine learning (ML) algorithms with CFD simulations could facilitate automated optimization of channel geometry and operational parameters. Similarly, the development of data-driven predictive models capable of estimating energy dissipation, turbulence intensity, and flow efficiency in real time would benefit hydraulic engineers in both design and operational decision-making. Finally, exploring multi-objective optimization frameworks, such as genetic algorithms (GA) or deep learning-based parameter tuning, could significantly improve the accuracy and efficiency of future Venturi channel designs.

5. Conclusions

This study comprehensively analyzed the hydraulic behavior of Venturi channels with three constriction ratios ($b/B = 0.35, 0.50$, and 0.65) using CFD simulations validated against experimental data. The numerical results showed strong agreement with experiments, with mean percentage errors of 4.12 % for velocity, 1.88 % for flow depth, and 7.36 % for the discharge coefficient, confirming the reliability of the simulations. Quantitative findings revealed that constriction geometry has a decisive impact on flow depth, velocity profiles, turbulence characteristics, and energy dissipation. Narrower constrictions ($b/B = 0.35$) increased flow depth and produced higher peak velocities and turbulent kinetic energy ($TKE = 0.028 \text{ m}^2/\text{s}^2$) compared to the wider geometry ($b/B = 0.65$, $TKE = 0.021 \text{ m}^2/\text{s}^2$). These effects resulted in nearly three-fold higher energy dissipation ($\Delta E/E_1 = 0.13$) and lower residual energy ratios ($E_2/E_1 = 0.87$) than the least constricted design ($\Delta E/E_1 = 0.05$, $E_2/E_1 = 0.95$). The narrower $b/B = 0.35$ geometry also produced a 24.42 % higher discharge coefficient compared to $b/B = 0.50$ and 18.71 % higher than $b/B = 0.65$, making it advantageous for applications requiring enhanced mixing or turbulence-driven processes. However, this comes at the cost of higher turbulence, longer recovery lengths, and reduced energy efficiency. Conversely, the $b/B = 0.65$ configuration consistently minimized energy dissipation, reduced turbulence intensity downstream, and promoted stable flow conditions, making it the most suitable option for applications where energy conservation and flow stability are critical. Energy dissipation analysis also indicated that at higher flow rates, narrower constrictions may achieve localized efficiency gains by accelerating discharge, while wider constrictions maintained stable performance across a broader range of flow conditions. These findings underscore the need to carefully balance constriction ratio and application requirements when designing Venturi channels. Highly constricted designs ($b/B = 0.35$) are recommended for processes where enhanced turbulence and mixing are desired (e.g., aeration, industrial mixing), while less constricted geometries ($b/B = 0.65$) are optimal for energy-efficient and stable flow applications (e.g., hydropower, sediment management, irrigation). Finally, the study demonstrates the utility of CFD as a robust tool for hydraulic design and provides a quantitative framework for optimizing Venturi channels. Future research should investigate transient flow conditions, sediment transport, non-Newtonian fluids, and hybrid configurations to further extend the applicability of these findings to complex real-world systems.

Funding

The authors declare that no funds, grants, or other support were received during the preparation of this manuscript.

Ethical approval

This article does not contain any studies with human participants or animals performed by any of the authors.

Informed consent

Informed consent was obtained from all individual participants included in the study.

CRedit authorship contribution statement

Amir Ghaderi: Writing – review & editing, Writing – original draft, Validation, Supervision, Software, Project administration, Methodology, Investigation, Data curation. **Amir Hossein Rezaei:** Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Data curation. **Fatemeh Nazari:** Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Formal analysis. **Hossein Mohammadnezhad:** Writing –

review & editing, Writing – original draft, Validation, Software, Methodology, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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